



OISTE/WISeKey ECHONET CP/CPS

Version 1.2.0

OISTE Policy Approval Authority

September 24, 2026

VERSION	DATE	MODIFICATION	AUTHOR
1.0	2026-01-09	First Version	Pedro Fuentes
1.1.0	2026-09-16	ECHONET-specific revision. Introduced the “ECHONET Requirements” framework (Appendix D); made the device certificate, revocation and repository provisions self-contained	Pedro Fuentes
1.2.0	2026-09-24	ECHONET Device Certificates issued with no scheduled expiration, as the Root and Subordinate CAs (section 6.3.2); retired Subordinate CAs are no longer revoked, revoked entries remain on the CRL permanently, and renewal, profile and retention provisions aligned accordingly	Pedro Fuentes

Contents

1. INTRODUCTION	9
1.1 Overview	9
1.2 Document name and identification	10
1.3 PKI participants	10
1.3.1 Certification authorities	10
1.3.2 Registration authorities	10
1.3.3 Subscribers (Requestors)	10
1.3.4 Relying parties	11
1.3.5 Other participants	11
1.4 Certificate usage	11
1.4.1 Appropriate certificate uses	11
1.4.2 Prohibited certificate uses	11
1.5 Policy administration	12
1.5.1 Organization administering the document	12
1.5.2 Contact person	12
1.5.3 Person determining CPS suitability for the policy	12
1.5.4 CPS approval procedures	12
1.6 Definitions and acronyms	12
2. PUBLICATION AND REPOSITORY RESPONSIBILITIES	12
2.1 Repositories	12
2.2 Publication of certification information	13
2.2.1 Statement on compliance with the ECHONET Requirements	13
2.3 Time or frequency of publication	13
2.4 Access controls on repositories	13
3. IDENTIFICATION AND AUTHENTICATION	13
3.1 Naming	13
3.1.1 Types of names	13
3.1.2 Need for names to be meaningful	14
3.1.3 Anonymity or pseudonymity of subscribers	14
3.1.4 Rules for interpreting various name forms	14
3.1.5 Uniqueness of names	14
3.1.6 Recognition, authentication, and role of trademarks	14
3.2 Initial identity validation	14
3.2.1 Method to prove possession of private key	14
3.2.2 Authentication of organization identity	14
3.2.3 Authentication of individual identity	15
3.2.4 Non-verified subscriber information	15
3.2.5 Validation of authority	15
3.2.6 Criteria for interoperation	15
3.3 Identification and authentication for re-key requests	15
3.3.1 Identification and authentication for routine re-key	16
3.3.2 Identification and authentication for re-key after revocation	16
3.4 Identification and authentication for revocation request	16
4. CERTIFICATE LIFE-CYCLE OPERATIONAL REQUIREMENTS	16
4.1 Certificate Application	16
4.1.1 Who can submit a certificate application	16
4.1.2 Enrollment process and responsibilities	16
4.2 Certificate application processing	17
4.2.1 Performing identification and authentication functions	17
4.2.2 Approval or rejection of certificate applications	17
4.2.3 Time to process certificate applications	17
4.3 Certificate issuance	17

4.3.1 CA actions during certificate issuance	17
4.3.2 Notification to subscriber by the CA of issuance of certificate	18
4.4 Certificate acceptance	18
4.4.1 Conduct constituting certificate acceptance	18
4.4.2 Publication of the certificate by the CA	18
4.4.3 Notification of certificate issuance by the CA to other entities	18
4.5 Key pair and certificate usage	18
4.5.1 Subscriber private key and certificate usage	18
4.5.2 Relying party public key and certificate usage	18
4.6 Certificate renewal	18
4.6.1 Circumstance for certificate renewal	19
4.6.2 Who may request renewal	19
4.6.3 Processing certificate renewal requests	19
4.6.4 Notification of new certificate issuance to subscriber	19
4.6.5 Conduct constituting acceptance of a renewal certificate	19
4.6.6 Publication of the renewal certificate by the CA	19
4.6.7 Notification of certificate issuance by the CA to other entities	19
4.7 Certificate re-key	19
4.7.1 Circumstance for certificate re-key	19
4.7.2 Who may request certification of a new public key	19
4.7.3 Processing certificate re-keying requests	19
4.7.4 Notification of new certificate issuance to subscriber	19
4.7.5 Conduct constituting acceptance of a re-keyed certificate	20
4.7.6 Publication of the re-keyed certificate by the CA	20
4.7.7 Notification of certificate issuance by the CA to other entities	20
4.8 Certificate modification	20
4.8.1 Circumstance for certificate modification	20
4.8.2 Who may request certificate modification	20
4.8.3 Processing certificate modification requests	20
4.8.4 Notification of new certificate issuance to subscriber	20
4.8.5 Conduct constituting acceptance of modified certificate	20
4.8.6 Publication of the modified certificate by the CA	20
4.8.7 Notification of certificate issuance by the CA to other entities	20
4.9 Certificate revocation and suspension	20
4.9.1 Circumstances for revocation	20
4.9.2 Who can request revocation	21
4.9.3 Procedure for revocation request	21
4.9.4 Revocation request grace period	22
4.9.5 Time within which CA must process the revocation request	22
4.9.6 Revocation checking requirement for relying parties	22
4.9.7 CRL issuance frequency	22
4.9.8 Maximum latency for CRLs	23
4.9.9 On-line revocation/status checking availability	23
4.9.10 On-line revocation checking requirements	23
4.9.11 Other forms of revocation advertisements available	23
4.9.12 Special requirements re key compromise	23
4.9.13 Circumstances for suspension	23
4.9.14 Who can request suspension	23
4.9.15 Procedure for suspension request	23
4.9.16 Limits on suspension period	23
4.10 Certificate status services	23
4.10.1 Operational characteristics	24
4.10.2 Service availability	24
4.10.3 Optional features	24
4.11 End of subscription	24
4.12 Key escrow and recovery	24

4.12.1 Key escrow and recovery policy and practices	24
4.12.2 Session key encapsulation and recovery policy and practices	24
5. FACILITY, MANAGEMENT, AND OPERATIONAL CONTROLS	24
5.1 Physical controls	25
5.1.1 Site location and construction	25
5.1.2 Physical access	25
5.1.3 Power and air conditioning	25
5.1.4 Water exposures	25
5.1.5 Fire prevention and protection	25
5.1.6 Media storage	25
5.1.7 Waste disposal	25
5.1.8 Off-site backup	25
5.2 Procedural controls	25
5.2.1 Trusted roles	26
5.2.2 Number of persons required per task	26
5.2.3 Identification and authentication for each role	26
5.2.4 Roles requiring separation of duties	26
5.3 Personnel controls	26
5.3.1 Qualifications, experience, and clearance requirements	26
5.3.2 Background check procedures	26
5.3.3 Training requirements	27
5.3.4 Retraining frequency and requirements	27
5.3.5 Job rotation frequency and sequence	27
5.3.6 Sanctions for unauthorized actions	27
5.3.7 Independent contractor requirements	27
5.3.8 Documentation supplied to personnel	27
5.4 Audit logging procedures	27
5.4.1 Types of events recorded	27
5.4.2 Frequency of processing log	28
5.4.3 Retention period for audit log	28
5.4.4 Protection of audit log	28
5.4.5 Audit log backup procedures	28
5.4.6 Audit collection system (internal vs. external)	28
5.4.7 Notification to event-causing subject	28
5.4.8 Vulnerability assessments	28
5.5 Records archival	28
5.5.1 Types of records archived	28
5.5.2 Retention period for archive	28
5.5.3 Protection of archive	29
5.5.4 Archive backup procedures	29
5.5.5 Requirements for time-stamping of records	29
5.5.6 Archive collection system (internal or external)	29
5.5.7 Procedures to obtain and verify archive information	29
5.6 Key changeover	29
5.7 Compromise and disaster recovery	29
5.7.1 Incident and compromise handling procedures	29
5.7.2 Computing resources, software, and/or data are corrupted	30
5.7.3 Entity private key compromise procedures	30
5.7.4 Business continuity capabilities after a disaster	30
5.8 CA or RA termination	30
6. TECHNICAL SECURITY CONTROLS	30
6.1 Key pair generation and installation	30
6.1.1 Key pair generation	31
6.1.2 Private key delivery to subscriber	31
6.1.3 Public key delivery to certificate issuer	31

6.1.4 CA public key delivery to relying parties	31
6.1.5 Key sizes	31
6.1.6 Public key parameters generation and quality checking	32
6.1.7 Key usage purposes (as per X.509 v3 key usage field)	32
6.2 Private Key Protection and Cryptographic Module Engineering Controls	32
6.2.1 Cryptographic module standards and controls	32
6.2.2 Private key (n out of m) multi-person control	32
6.2.3 Private key escrow	32
6.2.4 Private key backup	32
6.2.5 Private key archival	32
6.2.6 Private key transfer into or from a cryptographic module	32
6.2.7 Private key storage on cryptographic module	33
6.2.8 Method of activating private key	33
6.2.9 Method of deactivating private key	33
6.2.10 Method of destroying private key	33
6.2.11 Cryptographic Module Rating	33
6.3 Other aspects of key pair management	33
6.3.1 Public key archival	33
6.3.2 Certificate operational periods and key pair usage periods	34
6.4 Activation data	34
6.4.1 Activation data generation and installation	35
6.4.2 Activation data protection	35
6.4.3 Other aspects of activation data	35
6.5 Computer security controls	35
6.5.1 Specific computer security technical requirements	35
6.5.2 Computer security rating	35
6.6 Life cycle technical controls	35
6.6.1 System development controls	35
6.6.2 Security management controls	36
6.6.3 Life cycle security controls	36
6.7 Network security controls	36
6.8 Time-stamping	36
7. CERTIFICATE AND CRL PROFILES	36
7.1 Certificate profile	36
7.1.1 Version number(s)	36
7.1.2 Certificate extensions	37
7.1.3 Algorithm object identifiers	38
7.1.4 Name forms	38
7.1.5 Name constraints	39
7.1.6 Certificate policy object identifier	39
7.1.7 Usage of Policy Constraints extension	39
7.1.8 Policy qualifiers syntax and semantics	39
7.1.9 Processing semantics for the critical Certificate Policies extension	39
7.2 CRL profile	39
7.2.1 Version number(s)	39
7.2.2 CRL and CRL entry extensions	39
7.3 OCSP profile	39
8. COMPLIANCE AUDIT AND OTHER ASSESSMENTS	40
8.1 Frequency or circumstances of assessment	40
8.2 Identity/qualifications of assessor	40
8.3 Assessor's relationship to assessed entity	40
8.4 Topics covered by assessment	40
8.5 Actions taken as a result of deficiency	41
8.6 Communication of results	41

9. OTHER BUSINESS AND LEGAL MATTERS	41
9.1 Fees	41
9.1.1 Certificate issuance or renewal fees	41
9.1.2 Certificate access fees	41
9.1.3 Revocation or status information access fees	41
9.1.4 Fees for other services	41
9.1.5 Refund policy	41
9.2 Financial responsibility	42
9.2.1 Insurance coverage	42
9.2.2 Other assets	42
9.2.3 Insurance or warranty coverage for end-entities	42
9.3 Confidentiality of business information	42
9.3.1 Scope of confidential information	42
9.3.2 Information not within the scope of confidential information	42
9.3.3 Responsibility to protect confidential information	42
9.4 Privacy of personal information	43
9.4.1 Privacy plan	43
9.4.2 Information treated as private	43
9.4.3 Information not deemed private	43
9.4.4 Responsibility to protect private information	43
9.4.5 Notice and consent to use private information	43
9.4.6 Disclosure pursuant to judicial or administrative process	43
9.4.7 Other information disclosure circumstances	43
9.5 Intellectual property rights	43
9.6 Representations and warranties	44
9.6.1 CA representations and warranties	44
9.6.2 RA representations and warranties	44
9.6.3 Subscriber representations and warranties	44
9.6.4 Relying party representations and warranties	45
9.6.5 Representations and warranties of other participants	45
9.7 Disclaimers of warranties	45
9.8 Limitations of liability	45
9.9 Indemnities	45
9.10 Term and termination	45
9.10.1 Term	45
9.10.2 Termination	46
9.10.3 Effect of termination and survival	46
9.11 Individual notices and communications with participants	46
9.12 Amendments	46
9.12.1 Procedure for amendment	46
9.12.2 Notification mechanism and period	46
9.12.3 Circumstances under which OID must be changed	46
9.13 Dispute resolution provisions	47
9.14 Governing law	47
9.15 Compliance with applicable law	47
9.16 Miscellaneous provisions	47
9.16.1 Entire agreement	47
9.16.2 Assignment	47
9.16.3 Severability	47
9.16.4 Enforcement (attorneys' fees and waiver of rights)	47
9.16.5 Force Majeure	47
9.17 Other provisions	47
Appendix A: Glossary	47
Acronyms	47
Definitions	48

Appendix B: CA Hierarchies	50
ECHONET Root	50
Ownership and governance	50
Root Information	50
Subordinate CA Information	51
Appendix C: OID Inventory	51
Appendix D: Adopted ECHONET Requirements	51

1. INTRODUCTION

This document represents a combined Certificate Policy (CP) and Certification Practice Statement (CPS), and describes the practices followed with regard to the management of the lifecycle the Certification Authorities adhered to the OISTE/WISeKey Global Trust Model (OWGTM from now on) for the purposes of ECHONET devices.

1.1 Overview

The main two legal entities involved in the control and operation of the OISTE/WISeKey Global Trust Model are: - OISTE Foundation. The International Organization for Secure Electronic Transactions (“IOSET” or “OISTE”), a Swiss non-profit foundation established in 1998, and recognized with an “Special Consultative Status” by the United Nations. The OISTE Foundation maintains a Policy Approval Authority (PAA) that drafts, approves and revises the policies to which WISeKey is bound to comply with under its operator contract. The PAA is composed of members of the community to which OISTE provides its Certification Authority Services, resulting in a virtuous cycle for trust management. - WISeKey. WISeKey is referenced in this document as the short name for the entities “WISeKey International Holding Ltd.”, “WISeKey SA” or other members of the WISeKey Holding that are mandated by OISTE to host and operate the Root Certification Authorities and the technical infrastructures required to maintain the PKI at the appropriate operational level. WISeKey also operates as a “Subordinate Certification Authority” under the OISTE Roots, according to practices disclosed in this document.

The OISTE Global Trust Model (OWGTM) has been designed and are operated in accordance with the broad strategic direction of international PKI (Public Key Infrastructure) standards as well as their application to concrete identity frameworks in different domains (e.g. ID cards, passports, health cards, Internet of Things) and is intended to serve as a common Trust Model for Certification Authorities worldwide that comply with OISTE requirements.

The technologies, infrastructures, practices, and procedures implemented by the OWGTM have been designed with explicit standards of security in mind based on the requirements approved by OISTE.

The OISTE Foundation, under Swiss law, cannot belong to any individual or company. It is subject to annual supervision by the Swiss Federal Government and audited annually by independent auditors. Such supervision and audit require the foundation to pursue the objectives that have been set out for it, which includes the promotion of security in electronic communications worldwide.

This document is developed per the recommendations found in the document RFC3647, developed by the Internet Engineering Task Force (IETF), which has been adopted as a worldwide-recognized standard framework to document the Certifications Practice Statement and related Certificate Policies disclosed by a Certification Services Provider.

The purpose of this document is to disclose the Practices and Policies adopted in the OWGTM for the issuance of digital certificates. It is organized in the following sections: 1. Introduction – This section. Introduces the OWGTM and this document. 2. Publication and Repositories Responsibilities – Describes the publication policies for the certificates affected by this document, and the publication of this document itself. 3. Identification and Authentication – Discloses the rules for subscriber naming and required authentication policies. 4. Certificate Life-Cycle Operational Requirements – This section describes the different phases in the Life-Cycle of certificates and their requirements. 5. Management, Operational and Physical Controls – Describes the controls enforced in the OWGTM to provide adequate trust levels in the certificates issued under the Trust Model. 6. Technical Security Controls – Discloses the security controls adopted in the OWGTM. 7. Certificate and CRL Profiles – Describes the technical details of the different certificate types issued under the OWGTM. 8. Compliance Audit and other Assessment – Discloses the audit policies followed in the OWGTM to ensure that the participant fulfils the security and quality requirements. 9. Other Business and Legal Matters – This section exposes the commercial, legal and contractual aspects involved in the usage of certificates issued in the OWGTM.

APPLICABILITY NOTICE: If any inconsistency exists between this document and the normative provisions of an ECHONET Requirement (as defined in Appendix A and listed in Appendix D), then the ECHONET Requirement takes precedence over this CP/CPS, and the PAA shall amend this document at the next revision.

1.2 Document name and identification

Name	OISTE/WISeKey ECHONET Certificate Policy/Certification Practices Statement (CP/CPS)
Version	1.2.0
Issuance date	2026-09-24
Location	This document is also published in https://github.com/oiste/repository and https://wisekey.com/repository

1.3 PKI participants

The following sections describe the different participant types in the OWGTM.

1.3.1 Certification authorities

OISTE and WISeKey own and operate a number of Root and Issuing Certification Authority (CA) hierarchies that deliver certification services under the OWGTM.

The Root CA that anchors the ECHONET hierarchy is an exception to that ownership: it is owned by SealSQ Corp., while the trust model, policies and certification practices applied beneath it are regulated by the OISTE Foundation through its Policy Approval Authority. This is set out in Appendix B, where the hierarchy is detailed.

1.3.2 Registration authorities

The Registration Authorities are the physical or legal persons responsible for the identification of the entities requesting a certificate (referred as “applicants” when the request is in process and “subscribers” for those in possession of a certificate). The OWGTM delegates to Registration Authorities the responsibility of verifying the information provided by the applicant within a certificate request, ensuring that the request and the process used to deliver the certificate to the subscriber meets the requirements of this CPS and the appropriate CP.

The Registration Authorities in the OWGTM are directly supervised by the CA and follow an accreditation process imposed by the CA in order to ensure that all security and operational procedures related to the certificates life-cycle are strictly enforced. Within the OWGTM environment there exist locations named “OWGTM Registration Point” that are the physical or virtual locations where a Registration Authority operates. These Registration Points are operated by “Registration Authority Officers”, who are authorized persons responsible for verifying the identity and veracity of a certificate request for an end entity and the delivery of the certificate once issued by the Certification Authority.

Therefore, the responsibilities of Registration Authorities operating under the OWGTM are as follows: - Check the identity and circumstances needed to verify that a certificate request is valid according to the type of certificate requested. - Inform the applicant, before the issuance of the certificate, about the terms and conditions related to the certificate and its usage. - Verify that the information contained in a certificate is exact and complete according to the requirements of the corresponding CP. - Ensure that the subscriber is in possession of the digital signature creation data (private keys) associated to the certificate to be issued.

1.3.3 Subscribers (Requestors)

In the OWGTM two different end-user roles are defined. Depending on the status of the certificate request, these roles are named “Applicant” and “Subscriber”. In particular, for ECHONET Device Certificates the Subscriber is the Manufacturer, as stated in section 1.4.1 of this document. - An applicant is a physical person that requests a certificate for his own behalf or on behalf of a third party. The applicant needs to accredit his identity and ability to request a certificate. In the case of an applicant acting on behalf of a third party or legal person, he will be requested to accredit the empowerment for such representation, as required by law. - A subscriber is the physical or legal person whose identity is linked to the electronic signature creation data, or private key, and included in a digital certificate. In general, a subscriber is considered the

“owner” of a certificate. The subscriber of a certificate is responsible for the custody of his private key and not communicating this data in any way to any other person.

This document details the particular community of subscribers to whom each type of certificate is aimed and what identification and other security requirements should be fulfilled.

1.3.4 Relying parties

All natural and legal persons and other entities that trust the certificates issued by certification authorities operating under the OWGTM Trust Model are considered to be “relying parties”. These relying parties do not necessarily need to be a subscriber of an OWGTM certificate, but are requested to accept the “CertifyID Relying Party Agreement”, available at <http://oiste.org/repository>, which applies to relying parties of ECHONET Device Certificates.

In the OWGTM, a particular type of certificate could limit the right to be a relying party for that particular type of certificate, if this is the case, a specific Relying Party agreement would be published.

1.3.5 Other participants

No stipulation.

1.4 Certificate usage

In the OWGTM, the limitations for certificate usage are established for each particular certificate type. This information is summarized in the following subsections. The type of certificate is determined by the combination of “Key Usage”, “Extended Key Usage”, and Policy Identifiers.

In the context of ECHONET, certificate usage must be aligned with the ECHONET Requirements adopted by the PAA and listed in Appendix D.

1.4.1 Appropriate certificate uses

Certificate type	Description	Permitted uses
ECHONET Subordinate CA Certificate	Infrastructure certificate for a Subordinate Certification Authority authorised to issue ECHONET Device Certificates. It may be operated by WISeKey on behalf of a Manufacturer, or by the Manufacturer itself as a technically-constrained CA under section 3.2.6	Certificate Signing, CRL Signing
ECHONET Device Certificate	High-assurance device certificate used by an ECHONET device to authenticate itself and to protect transactions over ECHONET and other IoT networks. The certificate binds the device to its Manufacturer and to the manufacturer-assigned product and device identifiers, as verified under sections 3.2.2 and 3.2.3 and profiled in section 7.1	Digital Signature, Key Agreement, Client Authentication

1.4.2 Prohibited certificate uses

In general, any usage that is not explicitly stated in section 1.4.1 of this document or the appropriate CP, is considered to be prohibited.

1.5 Policy administration

This section describes how this document is administered. The same practices apply to all policies adopted by the OWGTM.

1.5.1 Organization administering the document

This document is administered by the OWGTM Policy Approval Authority (referred from now as PAA).

The PAA has a series of distinct functions but does not operate as a separate legal Entity. It is managed and organized in accordance with a process that draws on expertise within the OISTE Foundation and WISEKey. The PAA has been established to develop, review and/or approve the practices, policies and procedures for the entire Trust Model, subject to guidelines established by the members and advisors of the OISTE Foundation and WISEKey.

1.5.2 Contact person

- **Name:** OISTE Foundation - OWGTM Policy Approval Authority
- **email address:** cps@oiste.org, cps@wisekey.com
- **Address:** Avenue Louis-Casai 58 - 1216 Cointrin - Switzerland

This same contact can also be used for revocation requests and compliance-related notifications.

1.5.3 Person determining CPS suitability for the policy

The competent entity which determines the compliance and suitability of all CPS and the different supported CPs on behalf of the entire Trust Model is the OWGTM PAA.

1.5.4 CPS approval procedures

The OWGTM PAA defines and executes the procedures related to the approval of the CPS and CP and its subsequent amendments. Amendments will produce a new version of the document that will be published in the OWGTM Policy Repository (specified in section 2.1 of this document).

The approval of major changes of documents related to the PKI, and specially for the CP/CPS, require a meeting of the PAA and the issuance of an approval memo signed by at least two members of the PAA. Minor versions only require the participation of a single member of the PAA in order to approve the publication of a new version.

It's required to issue new CP/CPS versions at least once a year. In the case of versioning conflict, the latest version that prevails is always the document published in the Policy Repository.

Once any document of the Trust Model is updated, the CAs must do a technical assessment to identify any possible impact and/or required configuration changes in the platforms.

1.6 Definitions and acronyms

Definitions and Acronyms are included in Appendix A

2. PUBLICATION AND REPOSITORY RESPONSIBILITIES

The Root CA certificate and the Subordinate CA certificates of the ECHONET hierarchy are published in the OWGTM Policies repository identified in section 2.1, and are listed in Appendix B.

2.1 Repositories

The main repositories of the OWGTM are: - Policies repository for disclosure of CP/CPS and related information. This repository is a set of web pages and services available at the URLs <https://oiste.org> and <https://github.com/oiste/repository> - Certificate and Certificate Revocation information repositories. The CA certificates and Certificate Revocation Information sources are included, when relevant, as CDP and

AIA extensions in the certificates issued under the Root CA identified in Appendix B - Public Certificate repositories. The OWGTM makes ECHONET Root CA and Subordinate CA certificates, and the corresponding CRLs, publicly accessible to relying parties through the repository identified above. Should the ECHONET Consortium establish a certificate repository or ledger of its own, and the PAA adopt it as an ECHONET Requirement, it will be listed in Appendix D and used in addition to the OWGTM repository.

2.2 Publication of certification information

The OWGTM is responsible for publication of information regarding practices, certificates, and the current status of certificates. Where appropriate, such responsibilities may be delegated to the Subordinate CAs operating under the OISTE Trust Model.

The shared repositories containing public information in the OWGTM are managed by WISeKey SA or the operator of the Issuing CAs, and are available 24 hours a day, seven days a week. In the case of interruption by cause of “force majeure”, the service will be re-established in the minimum possible time.

2.2.1 Statement on compliance with the ECHONET Requirements

OISTE and WISeKey ensure compliance with industry best practices and security controls. In particular, the trust model enforces a regular review of the documents published by the ECHONET Consortium that are applicable to PKI services, certificate profiles or device identity. Those that the PAA formally adopts for the ECHONET hierarchy become “ECHONET Requirements” and are listed, with title, version and date of adoption, in Appendix D.

In the case of a discrepancy between any certification practice stated in this CP/CPS and an ECHONET Requirement, the ECHONET Requirement prevails, and the PAA shall amend this document at the next revision.

Where no ECHONET Requirement governs a given matter, this CP/CPS is self-contained and its own stipulations apply.

2.3 Time or frequency of publication

The CP/CPS documents will be published every time they are modified, with a minimum review period of one year. A certificate issued by any CA under the OWGTM will be published immediately after its issuance.

In the case of revocation of a certificate, the appropriate CA will include this revocation information in the Certificate Revocation Lists (CRL) according to section 4.9.7 (CRL issuance frequency).

2.4 Access controls on repositories

The OWGTM makes its Repository publicly available in a read-only manner.

3. IDENTIFICATION AND AUTHENTICATION

The OWGTM mandates the fulfillment of a set of required minimum controls that ensure the authenticity of the data included in certificates. These controls are enforced during the full lifecycle of certificates, certificate requests, and related documents

3.1 Naming

This section describes the elements regarding naming and identifying the subscribers of OWGTM certificates.

3.1.1 Types of names

All subscribers are assigned a Distinguished Name (DN) according to the X.501 Standard. This DN is composed of a Common Name (CN), which includes a unique identification of the subscriber as described in section 3.1.4.2, and a structure of X.501 components as defined in section 3.1.4.

3.1.2 Need for names to be meaningful

All Distinguished Names must be meaningful, and the identification the attributes associated to the subscriber should be in a human readable form.

3.1.3 Anonymity or pseudonymity of subscribers

ECHONET PKI CAs SHALL NOT issue anonymous or pseudonymous Certificates.

3.1.4 Rules for interpreting various name forms

The rules used in the OWGTM to interpret the distinguished names of certificates issued under its Trust Model are defined by the ISO/IEC 9595 (X.500) Distinguished Name (DN) standard.

3.1.5 Uniqueness of names

OWGTM requires uniqueness of names in the certificates issued by the Roots, except in the case of re-issuances or renewals for the same entity.

Multiple Certificates with the same subject name MAY be issued to the same Requestor. Name uniqueness is enforced for the entire Subject DN of the Certificate rather than an attribute (e.g., the common name). Each CA and RA SHALL identify the method for checking uniqueness of the Subject DNs within its domain.

3.1.6 Recognition, authentication, and role of trademarks

The inclusion of a name in a certificate does not imply any right over that name, neither for the OWGTM nor the applicant, nor the subscriber. The OWGTM reserves the right to refuse a certificate request, or revoke an existing one, if a conflict is detected over ownership or copyright of a name.

In any event, the OWGTM will not attempt to intermediate nor resolve conflicts regarding ownership of names or trademarks.

3.2 Initial identity validation

OWGTM performs “face to face” (or equivalent) identity validation for the certificates issued by the Roots. Stipulations related to subscriber certificates are defined in the following sections.

3.2.1 Method to prove possession of private key

If the key pair is generated by the End Entity (applicant or future subscriber), then a demonstration of the possession of the private key associated to the public key is requested. Accepted means are the generation of a Certificate Signing Request (CSR) linked to the private key, or any other method accepted by OWGTM.

If (when allowed by the applicable regulations) the key pair is generated by the CA or the RA, OWGTM defines and enforces approved procedures to transfer securely the private key to the subscriber (i.e. sending PFX files and passwords by different channels, and deleting any signature private key once the transfer is effective).

3.2.2 Authentication of organization identity

Before issuing a certificate for a subordinate Certification Authority OWGTM requires the fulfillment of a legally binding agreement between the organization and the OISTE Foundation, which includes the appropriate validation of the organization identity and signatories of the agreement.

Where a Manufacturer applies to operate its own ECHONET Subordinate CA, the OWGTM additionally verifies, before the Subordinate CA Certificate is issued, that the Manufacturer meets the accreditation criteria of section 3.2.6, that the Subordinate CA will be technically constrained to the Manufacturer Codes the Manufacturer is entitled to use, and that the Manufacturer accepts the audit obligations of section 8.4.

3.2.2.1 For Device Certificates

CP Identifier	Validation Policy
ECHONET Device Certificate	The Registration Authority must verify that:

3.2.3 Authentication of individual identity

The following subsections describe the required practices for each subscriber certificate type.

3.2.3.1 For Device Certificates

CP Identifier	Validation Policy
ECHONET Device Certificate	ID Data Verified:The Manufacturer Code (EPC 0x8A), the Product Code (EPC 0x8C) and the Production Number (EPC 0x8D) of the individual device, together with the legal name and country of the Manufacturer. Method of Verification:The Manufacturer Code and the Manufacturer identity are verified as stated in section 3.2.2. The Product Code and Production Number are taken from the Manufacturer's production record for the unit concerned; the CA accepts them on the basis of the Manufacturer's warranty in the Subscriber Agreement that each Production Number is unique within the Product Code and identifies a single physical unit. Enrolment requests must originate from a production system authenticated to the Issuing CA or Subordinate CA, and must be constrained to the Manufacturer Codes assigned to that Manufacturer. Entities authorized to verify:

3.2.4 Non-verified subscriber information

All attributes included in a certificate that are subject to the ECHONET Requirements or to this CP/CPS must undergo appropriate validation. No unverified subscriber information is included in ECHONET Device Certificates.

3.2.5 Validation of authority

The CA's Certificate issuance process SHALL confirm that: - The corporate contact listed in the Registration Authority Declaration (the signed enrolment record establishing the organization's authorised contacts, "RAD") is an officer of the organization who can sign on its behalf and bind it to the terms and conditions of the agreement; - The representative submitting the RAD and the Certificate Application is authorized to act on behalf of the organization; - The administrators listed in the RAD are authorized to act on behalf of the organization; and - The contacts listed in the RAD are authorized to act on behalf of the organization.

3.2.6 Criteria for interoperation

A Certification Authority that wishes to interoperate with the OWGTM is required to undergo an internal accreditation process to ensure the compliance with this CPS.

If this accreditation process is successful, it will result in the creation of an "Issuing CA" under the OWGTM that adheres to this CPS and authorized to issue certain Certificate Types.

3.3 Identification and authentication for re-key requests

This section addresses the following elements for the identification and authentication procedures for re-key for each subject type (CA, RA, subscriber, and other participants). Unless otherwise specified, it can

considered as equivalent the activities linked to “re-key” (new certificate for an existing subscriber, using a new key pair) and “renewal” (new certificate for an existing subscriber, using the same key pair).

3.3.1 Identification and authentication for routine re-key

The certificate subscriber can request a routine re-key by authenticating himself with one of these methods: - Username & Password - A valid digital certificate linked to the user account

3.3.2 Identification and authentication for re-key after revocation

The OWGTM does not support re-key of certificates after revocation. The subscriber must apply for a new digital certificate by using the same procedures as for its issuance.

3.4 Identification and authentication for revocation request

The Identification Policy for revocation requests is, generally, the same as stipulated for initial registration. The preferred method to authenticate revocation requests is an authentication based in a digital certificate owned by the certificate subscriber, or authorized party. Passwords maybe accepted alternatively.

A Certification Authority may define, that during the enrolment process, a subscriber can create a password that can be used in remote revocation requests, using an on-line procedure communicated to the user when issuing the certificate.

4. CERTIFICATE LIFE-CYCLE OPERATIONAL REQUIREMENTS

The stipulations included in this section are understood as common for all the certificates issued under the OWGTM Root, unless otherwise specified in this document.

When applicable, CAs operating under the OWGTM must respect the ECHONET Requirements.

4.1 Certificate Application

For CA Certificates, before issuing a new certificate for a subordinate Certification Authority OWGTM requires the fulfillment of a legally binding agreement between the affiliated organization and the OISTE Foundation, which includes the appropriate validation of the organization identity and signatories of the agreement. Additionally, for each Subordinate CA, it's required the fulfillment of a “Subordinate CA Naming Application”, which must be signed by an authorized representative of the affiliate.

For subscriber certificates, the Registration Authorities operating under the OWGTM are competent and responsible for determining if the type of the requested certificate is adequate for the applicant and future subscriber, in conformity with the Certificate Policy related to that certificate, and therefore to proceed or not with the certificate application. The Certificate Application process must include a mean to express acceptance with the Subscriber Agreement, by means of a manuscript signature or another valid mechanism, and it's a first step to begin the certificate issuance process.

4.1.1 Who can submit a certificate application

A certificate application can be submitted by the subject of the certificate or by an authorized representative of the subject.

4.1.2 Enrollment process and responsibilities

WISeKey is responsible for ensuring that the identity of each Certificate Applicant is verified in accordance with this CP and the applicable CPS prior to the issuance of a Certificate. Applicants are responsible for submitting sufficient information and documentation for the Issuer CA or the RA to perform the required verification of identity prior to issuing a Certificate.

This process includes the identification of suspicious or potentially dangerous requests, based on automated checks against the list of Manufacturers whose enrolment rights have been suspended or withdrawn, against previously revoked certificates, and against previously denied requests marked as suspicious.

In particular and where applicable, CAs will respect the ECHONET Requirements.

4.2 Certificate application processing

This section describes the procedures for processing certificate applications in the OWGTM Trust Model.

4.2.1 Performing identification and authentication functions

Before issuing a certificate from the Root CA identified in Appendix B for a subordinate Certification Authority, it's required that two representatives of the PAA verify the Subordinate CA Naming Application and the rightfulness to operate a subordinate CA under that Root.

The identification and authentication functions are delegated to the Registration Authorities operating under the OWGTM.

An authorized Registration Authority Officer will perform these functions. This role can be assumed by:

- An accredited person that, on behalf of a Registration Authority, personally executes the identification and authentication functions.
- An accredited software application that performs the identification and authentication functions for automated certification procedures. Automated registration is expressly permitted for ECHONET Device Certificates, where enrolment occurs in a manufacturing environment under a Manufacturer whose identity and identifiers have already been validated under sections 3.2.2 and 3.2.3. Any accredited software application will execute this function according to sections 3.2.2 and 3.2.3 of this document.

The steps to be executed by the Issuing CA or RA are as follows:

- As a first step, the Issuing CA or RA will perform the verifications stipulated in section 3.2.
- As a second step, the Issuing CA must check the certificate details against a list of previously revoked Certificates and rejected certificate requests to identify suspicious certificate requests.

The Issuing CA can only issue a certificate after having successfully completed the above steps.

4.2.2 Approval or rejection of certificate applications

An approval of a certificate application derives from the execution of the certificate issuance procedures, as defined in the section 4.3 of this document.

A rejection of a certificate application results in a notification being sent to the applicant by appropriate means and is registered for further reference.

4.2.3 Time to process certificate applications

There is no time limit stipulated to complete the processing of an application.

4.3 Certificate issuance

A certificate request will be forwarded to a Certification Authority for its issuance only after the Registration Authority confirms the correctness of the information contained in the request. The OWGTM is not responsible for monitoring, research or confirmation of the correctness of the information contained in a certificate during the intermediate period between its issuance and renewal, unless this period exceeds the re-verification limits stipulated in section 3.3 of this CP/CPS.

4.3.1 CA actions during certificate issuance

A Certification Authority adhering to the OWGTM proceeds with the issuance of a certificate only after executing the necessary measures to verify that the signing request is authorized and genuine, as per the particular controls are stipulated in this document.

4.3.2 Notification to subscriber by the CA of issuance of certificate

For CA Certificates, OWGTM notifies directly to the authorized CA responsible.

WiSeKey will send, in general, all notifications to the subscriber using email to the address specified in the application process. These notifications should include a digital signature.

4.4 Certificate acceptance

Certificate acceptance is the final step in the certification issuance process. After Acceptance the certificate owner is entitled to use the certificate and issue valid digital signatures.

4.4.1 Conduct constituting certificate acceptance

For CA Certificates the CA representative must acknowledge the reception of the certificate, verifying that the Key Fingerprint matches the request. Installing the CA Certificate in the CA server constitutes tacit acceptance.

Certificate acceptance is understood after the subscriber or his representative performs one or more of the following: - Accepts the “Subscriber Agreement”, which includes the terms and conditions associated with the particular Certificate Policy, and which constitutes formal acceptance of those terms; or - Downloads and/or installs the certificate, making it technically available for usage; or - Doesn’t expressly refuse the certificate once the issuance notification has been sent.

4.4.2 Publication of the certificate by the CA

The CAs operating under the OWGTM publish all issued certificates as specified in section 2 of this document.

4.4.3 Notification of certificate issuance by the CA to other entities

The CA only notifies the Registration Authority from which it received the request of the issuance of a certificate. It is the RA’s duty to notify the certificate subscriber, as stipulated in section 4.3.2 of this CPS.

4.5 Key pair and certificate usage

The certificates issued by the OWGTM are used to provide authenticity, integrity, confidentiality and/or non-repudiation in electronic transactions and other computerized functions.

4.5.1 Subscriber private key and certificate usage

For CA Certificates the private key may only be used according to the CPS published by the subordinate CA, subject to approval by the OWGTM PAA.

The specific usages allowed for a private key associated to a certificate type issued in the OWGTM are as summarized in section 1.4 of this document

4.5.2 Relying party public key and certificate usage

Relying parties must access and use the public key and certificates issued under the OWGTM as stipulated in this CPS and as indicated in the “Relying Party Agreement” document, made public at the web page <http://www.oiste.org/repository>.

4.6 Certificate renewal

Certificate Renewal is understood as the issuance of a new certificate to a subscriber who maintains the key pair generated for the original certificate. Certificate renewal may not be supported depending on business decisions.

4.6.1 Circumstance for certificate renewal

For CA Certificates it is allowed the certificate renewal for the purpose of modifying the name constraints, enhanced key usages or other non-identity extensions. Renewal is not used to extend a validity period, no CA certificate in the ECHONET hierarchy having a scheduled expiration (section 6.3.2).

For Subscriber Certificates issued without a scheduled expiration, renewal does not apply. Where a Manufacturer has requested a fixed validity for a product line under section 6.3.2, certificate renewal is allowed for the purpose of extending the validity period, always considering the requirements for re-verification periods stipulated in section 3.3 of this CPS.

4.6.2 Who may request renewal

The certificate renewal can be requested by the same entities allowed to request the first issuance of the certificate.

4.6.3 Processing certificate renewal requests

Certificate renewal requests are processed according to the same rules than the initial issuance.

4.6.4 Notification of new certificate issuance to subscriber

The notification of the issuance of a renewed certificate it will occur as described in section 4.3.2 of this document.

4.6.5 Conduct constituting acceptance of a renewal certificate

As stipulated in section 4.4.1 of this document.

4.6.6 Publication of the renewal certificate by the CA

The CAs operating under the OWGTM publish all issued certificates as specified in section 2 of this document.

4.6.7 Notification of certificate issuance by the CA to other entities

The CA only notifies the Registration Authority from which it received the request of the issuance of a certificate. It is the RA's duty to notify the certificate subscriber, as stipulated in section 4.3.2 of this document.

4.7 Certificate re-key

Certificate Re-Key is understood as the issuance of a new certificate to a subscriber that also generates a new key pair. This process is supported for all certificate types.

4.7.1 Circumstance for certificate re-key

Any certificate that is not revoked can be re-keyed.

4.7.2 Who may request certification of a new public key

The certificate renewal can be requested by the same entities allowed to request the first issuance of the certificate.

4.7.3 Processing certificate re-keying requests

Certificate re-key requests are processed according to the same rules than the initial issuance.

4.7.4 Notification of new certificate issuance to subscriber

The notification of the issuance of a new certificate it will occur as described in section 4.3.2 of this document.

4.7.5 Conduct constituting acceptance of a re-keyed certificate

As stipulated in section 4.4.1 of this document.

4.7.6 Publication of the re-keyed certificate by the CA

The CAs operating under the OWGTM publish all issued certificates as specified in section 2 of this document.

4.7.7 Notification of certificate issuance by the CA to other entities

The CA only notifies the Registration Authority from which it received the request of the issuance of a certificate. It is the RA's duty to notify the certificate subscriber, as stipulated in section 4.3.2 of this document.

4.8 Certificate modification

The OWGTM does not allow the modification of certificates during their validity period. If the information contained in a certificate cease to be valid, or the circumstances of the subscriber change in such a manner that the conditions expressed in the CPS or the CP are not met, then the only accepted procedure is the revocation and re-issuance of a new certificate.

4.8.1 Circumstance for certificate modification

No stipulation. Modification is not allowed.

4.8.2 Who may request certificate modification

No stipulation. Modification is not allowed.

4.8.3 Processing certificate modification requests

No stipulation. Modification is not allowed.

4.8.4 Notification of new certificate issuance to subscriber

No stipulation. Modification is not allowed.

4.8.5 Conduct constituting acceptance of modified certificate

No stipulation. Modification is not allowed.

4.8.6 Publication of the modified certificate by the CA

No stipulation. Modification is not allowed.

4.8.7 Notification of certificate issuance by the CA to other entities

No stipulation. Modification is not allowed.

4.9 Certificate revocation and suspension

All Certification Authorities operating under the OWGTM ensure, by establishing the necessary means, that a certificate that compromises the Trust Model for any reason is prevented from being used by either revoking or suspending that certificate.

Suspension of certificates is not supported.

4.9.1 Circumstances for revocation

All certificate subscribers receiving a digital certificate issued under a Root regulated by this CPS must assume the stipulations contained in this section.

4.9.1.1 Reasons for Revoking a Subscriber Certificate A Certification Authority operating in the OWGTM must revoke a certificate that it has issued upon the occurrence of any of the following events: 1. The Subscriber requests in writing that the CA revoke the Certificate; 2. The Subscriber notifies the CA that the original certificate request was not authorized and does not retroactively grant authorization; 3. The CA obtains evidence that the Subscriber's Private Key corresponding to the Public Key in the Certificate suffered a Key Compromise; 4. The CA is made aware of a demonstrated or proven method that can easily compute the Subscriber's Private Key based on the Public Key in the Certificate; or 5. The CA obtains evidence that the validation of the Manufacturer identity, or of any device identifier included in the Certificate, should not be relied upon.

A Certification Authority operating in the OWGTM must revoke a certificate that it has issued upon the occurrence of any of the following events: 1. The Certificate no longer complies with the requirements of Sections 6.1.5 and 6.1.6; 2. The CA obtains evidence that the Certificate was misused; 3. The CA is made aware that a Subscriber has violated one or more of its material obligations under the Subscriber Agreement or Terms of Use; 4. The CA is made aware that the Manufacturer is no longer entitled to use a Manufacturer Code, product identifier or other identifier included in the Certificate; 5. The CA is made aware of a material change in the information contained in the Certificate; 6. The CA is made aware that the Certificate was not issued in accordance with this CP/CPS or the CA's Certificate Policy or Certification Practice Statement; 7. The CA determines or is made aware that any of the information appearing in the Certificate is inaccurate; 8. The CA's right to issue Certificates under this CP/CPS expires or is revoked or terminated, unless the CA has made arrangements to continue maintaining the CRL Repository; 9. Revocation is required by the CA's Certificate Policy and/or Certification Practice Statement; or 10. The CA is made aware of a demonstrated or proven method that exposes the Subscriber's Private Key to compromise, methods have been developed that can easily calculate it based on the Public Key, or if there is clear evidence that the specific method used to generate the Private Key was flawed.

Revocation of ECHONET Certificates: revocation is processed as stated in this section 4.9 and, where an ECHONET Requirement stipulates additional or shorter deadlines, as stated in that ECHONET Requirement.

4.9.1.2 Reasons for Revoking a Subordinate CA Certificate An issuing Certification Authority operating in the OWGTM will be revoked upon the occurrence of any of the following events: 1. The Subordinate CA requests revocation in writing; 2. The Subordinate CA notifies the Issuing CA that the original certificate request was not authorized and does not retroactively grant authorization; 1. The Issuing CA obtains evidence that the Subordinate CA's Private Key corresponding to the Public Key in the Certificate suffered a Key Compromise or no longer complies with the requirements of Sections 6.1.5 and 6.1.6; 1. The Issuing CA obtains evidence that the Certificate was misused; 2. The Issuing CA is made aware that the Certificate was not issued in accordance with or that Subordinate CA has not complied with this document or the applicable Certificate Policy or Certification Practice Statement; 1. The Issuing CA determines that any of the information appearing in the Certificate is inaccurate or misleading; 1. The Issuing CA or Subordinate CA ceases operations for any reason and has not made arrangements for another CA to provide revocation support for the Certificate; 1. The Issuing CA's or Subordinate CA's right to issue Certificates under this CP/CPS expires or is revoked or terminated, unless the Issuing CA has made arrangements to continue maintaining the CRL Repository; 1. Revocation is required by the Issuing CA's Certificate Policy and/or Certification Practice Statement; or 1. Revocation is required by the OISTE Foundation.

4.9.2 Who can request revocation

The certificate subscriber or its legal representative can request the revocation of an individual or organizational certificate.

Third parties may request certificate revocation for problems related to fraud, misuse, or compromise. Certificate revocation requests must identify the entity requesting revocation and specify the reason for revocation.

4.9.3 Procedure for revocation request

The procedure to be used for certificate revocation requests is detailed in the "ECHONET Certificate Subscriber Agreement", published in the repository identified in section 2.1. Manufacturers enrolling and managing

ECHONET Device Certificates through the INeS IoT Security Portal can request revocation through that same service.

To report suspected Private Key Compromise, Certificate misuse, Certificate mis-issuance, or other types of fraud, compromise, misuse, inappropriate conduct, or any other matter related to Certificates, the main and preferred method is sending an e-mail message to cps@wisekey.com.

For certificate subscribers that seek to obtain general support, the preferred method to communicate with WISEKey is sending an e-mail message to support@wisekey.com.

The common practice for all certificates issued under the OWGTM Trust Model is for revocation requests to be accepted automatically and produce an immediate revocation in the case of: - Remote requests sent by e-mail or via a web page or service, appropriately authenticated by the subscriber or its representative. - Face-to-face requests addressed to an official Registration Authority representative and the identity of the requestor is proved by the same means as used for certificate registration. - Revocation requests sent by an official Registration or Certification representative operating in the OWGTM.

4.9.4 Revocation request grace period

There is no stipulation for grace periods for revocation requests. The revocation process will be started immediately upon the receipt of such a request by an authorized party.

4.9.5 Time within which CA must process the revocation request

Revocation requests are processed by the CA within the shortest possible period, and always in accordance with the limits set in section 4.9.1.

The CA begins investigating a Certificate Problem Report within 24 hours of receipt. Following that investigation, the CA revokes the Certificate within the following maximum periods:

Circumstance	Maximum period to revoke
Key Compromise of the Subscriber Private Key, or of a Subordinate CA Private Key	24 hours
The Certificate was mis-issued, or the CA obtains evidence that it was misused	24 hours
Any other circumstance listed in section 4.9.1	5 days

Where an ECHONET Requirement stipulates a shorter period, that shorter period applies.

4.9.6 Revocation checking requirement for relying parties

The OWGTM requires that all parties willing to rely on certificates issued under the Trust Model check the status of these Certificates on each digital signature verification and authentication request using the certificate. For certificates issued under the ECHONET hierarchy this requirement is fulfilled by consulting the most recent CRL published by the CA that issued the Certificate. No online status protocol is offered for these certificates.

The information necessary to locate the CRL is included in all ECHONET certificates, using the standard `CRLDistributionPoints` extension.

4.9.7 CRL issuance frequency

The OISTE CAs used by the OWGTM issue a full CRL at least every year, with a typical overlapping period of one week. This CRL will contain the revoked, if any, certificates for OWGTM Policy CAs or Issuing CAs, as appropriate for the hierarchy. New CRLs are published immediately if a new subordinated CA is revoked.

Subordinate Certification Authorities issuing ECHONET Device Certificates issue a full CRL at least every 12 months, and in any case within 24 hours of a revocation being recorded. The `nextUpdate` field of a CRL is set no more than 12 months beyond its `thisUpdate` field.

4.9.8 Maximum latency for CRLs

CRLs are posted to their distribution point within the minimum possible time after generation.

4.9.9 On-line revocation/status checking availability

On-line status checking by the Online Certificate Status Protocol is not offered for certificates issued under the ECHONET hierarchy. Revocation status is published exclusively by CRL, as stated in sections 4.9.7 and 4.10.

4.9.10 On-line revocation checking requirements

Not applicable. No on-line status checking protocol is offered.

Relying parties are requested to always check the validity of the certificate on which they rely, by the means stipulated in section 4.9.6.

4.9.11 Other forms of revocation advertisements available

No stipulations.

4.9.12 Special requirements re key compromise

Any party detecting a key compromise at any level in the OWGTM Trust Model is requested to immediately communicate it to a Registration or Certification Authority.

In particular for ECHONET certificates, but applicable for any other certificate type issued, it's also requested to Subscribers, Relying Parties, Application Software Vendors and other third parties to report any potential issue to the Certification Authority (Certificate misuse, or other types of fraud, compromise, misuse, or inappropriate conduct related to Certificates).

The appropriate methods to demonstrate key compromise are: - Create and sign a text file, - Create a custom CSR file, and/or - Send the private key, or a link to where it's publicly disclosed.

The main method for these communications is the stipulated in section 4.9.3.

4.9.13 Circumstances for suspension

Suspension is not allowed for any certificate in scope of this CPS.

4.9.14 Who can request suspension

No stipulation. Suspension is not supported for any certificate in scope of this CP/CPS.

4.9.15 Procedure for suspension request

No stipulation. Suspension is not supported for any certificate in scope of this CP/CPS.

4.9.16 Limits on suspension period

No stipulation. Suspension is not supported for any certificate in scope of this CP/CPS.

4.10 Certificate status services

Any CA operating in the OWGTM must provide a highly available and reliable service for checking the status of all certificates issued under its Trust Model.

4.10.1 Operational characteristics

Certificate Status Services are accessible through HTTP servers owned by the OWGTM Certification Authorities, and are accessed by downloading the applicable Certificate Revocation List (CRL). CRLs are served over HTTP without authentication or access restriction.

The applicable CRL URL is included in the cRLDistributionPoints extension of every certificate issued under the ECHONET hierarchy.

4.10.2 Service availability

The Certificate Status Services are available on a 24x7 basis.

4.10.3 Optional features

No stipulation.

4.11 End of subscription

“End of Subscription” is understood to occur after the expiration or revocation of a certificate, and it is unique for that particular certificate, not affecting additional subscriptions (if any) that the end entity may hold within the OWGTM.

4.12 Key escrow and recovery

Key escrow is not permitted for ECHONET Certificates.

4.12.1 Key escrow and recovery policy and practices

Not applicable. Key escrow is not offered for any certificate in scope of this CP/CPS.

4.12.2 Session key encapsulation and recovery policy and practices

No stipulation.

5. FACILITY, MANAGEMENT, AND OPERATIONAL CONTROLS

This section describes the non-technical security controls used by the participants involved in the issuance, publishing and management of keys within the OWGTM. The OWGTM asserts the importance of these controls as a fundamental basis to provide trust to subscribers and all relying parties, and therefore establishes and maintains the necessary means to ensure and demonstrate that these controls are enforced.

These controls are under surveillance and audited both internally and externally by accredited bodies. The public manifests of these audits are published on a regular basis in the OWGTM web site (<http://www.oiste.org/repository>).

The OWGTM allows third parties to host and operate some of the components of its infrastructure. If such a delegation occurs, the assigned party will be requested to meet the controls stipulated in this section and an auditing process will be executed to ensure that the necessary measures to ensure these controls are effective are in place and enforced.

In particular: - The OISTE Foundation delegates the hosting and operations of the “Root CA” and the “Policy CAs” (and related certificate publication and verification services) to WISEKey. - The “Issuing CAs” (and related certificate publication and verification services) are hosted and operated by WISEKey (except for the cases of technically-constrained CAs, which could be hosted by their owners). These participants are allowed to delegate the hosting and operation to WISEKey only; other delegations or outsourcing are only permitted after a security assessment and a formal authorization. - Registration Authorities and Registration Authority Points are appointed by WISEKey. Registration Authorities are not allowed to delegate their operations to other parties without the approval and direct supervision of WISEKey.

5.1 Physical controls

This section describes the physical controls on facilities housing OWGTM components.

5.1.1 Site location and construction

The OWGTM information systems are located in Secure Datacenters providing adequate security levels and under surveillance 24 hours a day, 7 days a week. These Datacenters are built in such a manner that relevant critical physical risks are managed.

5.1.2 Physical access

The OWGTM Secure Datacenter implements diverse nested security perimeters. The access from an outer to an inner perimeter requires different security and authorization controls. Among these controls, biometric door access, video surveillance and intrusion detection systems are implemented.

5.1.3 Power and air conditioning

The OWGTM Secure Datacenter implements power and air conditioning systems sufficiently dimensioned to accommodate the operating needs.

5.1.4 Water exposures

The facilities are located in a place where natural flooding risks are controlled, and they are equipped with flooding sensors and alarms.

5.1.5 Fire prevention and protection

The facilities implement fire detection, prevention and protection controls.

5.1.6 Media storage

Sensible information media are stored securely in fireproof containers and high security safes, depending on the media type and the classification of the information they contain.

These containers and safes are located in redundant placements, in order to eliminate the risks of using a single location (i.e. in the case of fire or water damage).

Access to these storage locations and items is restricted to authorized persons and regulated by security procedures.

5.1.7 Waste disposal

The disposal of optical or magnetic media and paper containing any information generated during OWGTM operations is executed following procedures established for such purposes, including demagnetization and/or destruction processes, depending on the media type to be disposed.

5.1.8 Off-site backup

OWGTM executes a backup copy of all information needed to promote a secondary datacenter to operational status in the event of a disaster preventing the main datacenter from maintaining an adequate service level.

A remote backup copy is periodically made and stored in a way such that dual access control is required to restore the backup copies.

5.2 Procedural controls

The information systems and services incorporated in the OWGTM are operated in a secure manner, following a set of predefined procedures that are enforced by the OWGTM and verified through periodical auditing activities.

For security reasons the information related to these controls are classified as “CONFIDENTIAL” and this document may only disclose a summarized version. Further detailed information is only disclosed to accredited auditors who are responsible for reviewing OWGTM components and operations.

5.2.1 Trusted roles

The OWGTM establishes and enforces a strict security policy to control all operations performed at any level of the Trust Model. This includes the identification and control of the Persons performing those operations. These Persons are considered “Trusted Roles” and include, but are not limited to: - Certification Authority Administrator - Certification Authority Operator - Registration Authority Administrator - Registration Authority Operator with Vetting attributions - Registration Authority Operator without Vetting attributions - Systems Administrator - Security Administrator - Policy Approval Authority Member

Persons seeking to become Trusted Persons by obtaining a Trusted Position must successfully complete the screening requirements set out in this CPS (section 5.3).

5.2.2 Number of persons required per task

The OWGTM establishes the need for the segregation of duties based on job responsibility in order to ensure that the adequate number of Trusted Persons is required to perform sensitive tasks.

The roles requiring separation of duties is stipulated in section 5.2.4.

5.2.3 Identification and authentication for each role

All the persons assuming a role in the OWGTM systems follow an authorization process that entitles them to access the appropriate information and systems for their role.

Physical access control for all the authorized persons accessing OWGTM’s systems and services systems is typically enforced using two factor authentication that usually includes biometrics.

5.2.4 Roles requiring separation of duties

Roles requiring Separation of duties include at least the following: - Any activity involved in the operation of a Root Certification Authority. - Enabling a CA into a production status (CA Ceremony procedures) - Issuance, or revocation of CA Certificates - Validation of information and issuance of ECHONET Subordinate CA Certificates, and the authorisation of a Manufacturer to enrol ECHONET Device Certificates

5.3 Personnel controls

Personnel bearing one of the roles defined in section 5.2.1 will be required to fulfil the “OWGTM Trusted Professional Policy”, summarized in the following sections.

5.3.1 Qualifications, experience, and clearance requirements

Personnel acting directly or indirectly for the OWGTM will be required to possess the required qualification and/or proved experience in certification service provision environments. All involved personnel will be required to act according to the OWGTM Security Policy and to possess: - Knowledge and training (according to the role assigned to the person) in Public Key Infrastructures. - Knowledge and training (according to the role) in Information Systems Security. - Knowledge and training specific for the responsibilities assigned.

5.3.2 Background check procedures

The Human Resource Department conducts verification checks on permanent staff at the time of job applications, and ensures that all personnel with access to sensitive information are trustworthy and understand their responsibilities; this includes at a minimum the following: - Availability and verification of satisfactory references; - Confirmation of claimed academic and professional qualifications; - Identity checks of passport or similar document.

5.3.3 Training requirements

Personnel directly involved in OWGTM, including “Issuing CAs” operated by third parties and Registration Authorities, will follow an internal training plan adapted to their assigned attributions. This training will be compliant with industry regulations and with the ECHONET Requirements, as applicable.

5.3.4 Retraining frequency and requirements

Retraining sessions are required for all involved personnel in the case of environmental, technology and/or operative changes. Changes in practices and/or policies are communicated to all involved personnel.

5.3.5 Job rotation frequency and sequence

No stipulation.

5.3.6 Sanctions for unauthorized actions

If an unauthorized action is detected the OWGTM will undertake necessary disciplinary actions. Any action that (intentionally or unintentionally) contravenes the Certification Practice Statement.

Upon detection of an unauthorized action the OWGTM will initiate an investigation process. During this process the involved persons will be prevented from obtaining access to OWGTM systems and information.

Disciplinary actions will be taken after the investigation determines the severity and intent of the action.

5.3.7 Independent contractor requirements

External contractors are required to agree with the Information Security policies of the OWGTM and temporary staff not already covered by an existing confidentiality agreement shall also be required to sign the Non-Disclosure Agreement prior to being granted access to Information resources.

The agreement is reviewed when there are changes to employment terms or contracts.

5.3.8 Documentation supplied to personnel

All personnel incorporated within the OWGTM are provided access, as required for their role, to the following information: - Certification Practices Statement - Certificate Policies - Privacy Policy - Security Policy - Organization chart and assigned functions and responsibilities - Operational procedures - Incident response procedures

5.4 Audit logging procedures

This section describes the event logging and audit systems that have been implemented to maintain a secure environment in the OWGTM.

5.4.1 Types of events recorded

OWGTM records in their servers all events related to: - CA key lifecycle management events, including: 1. Key generation, backup, storage, recovery, archival, and destruction as captured by procedure documentation; and 2. Cryptographic device lifecycle management events as captured by procedure documentation. - CA and Subscriber Certificate lifecycle management events, limited to: 1. Certificate requests and revocation as captured by CA logs; 2. Verification activities 3. Date, time, phone number used, persons spoken to, and end results of verification telephone calls as captured by registration officers; 4. Acceptance and rejection of certificate requests as captured by CA logs; 5. Issuance of Certificates as captured by CA logs 6. Generation of Certificate Revocation Lists as may be captured by CA logs (NB CRLs are not retained, only the record of its generation) - Security events, including: 1. Successful and unsuccessful PKI system access attempts as captured by operating system logs; 2. Major PKI and security system actions performed as captured by operational logs; 3. Security profile changes as captured by operating system logs; 4. System crashes, hardware failures, and other anomalies in server logs; 5. Entries to and exits from the CA facility as captured by access control logs. - Router and Firewall Activities Logs 1. Successful and unsuccessful login attempts

to routers and firewalls; and 2. Logging of all administrative actions performed on routers and firewalls, including configuration changes, firmware updates, and access control modifications; and 3. Logging of all changes made to firewall rules, including additions, modifications, and deletions; and 4. Logging of all system events and errors, including hardware failures, software crashes, and system restarts.

5.4.2 Frequency of processing log

Logs are processed and audited when required.

For systems that are kept offline, as the Root CA, audit logs are only reviewed when an operation is executed.

5.4.3 Retention period for audit log

OWGTM and involved parties retain all audit logs as specified in section 5.5.2.

5.4.4 Protection of audit log

All audit records and archives are stored in fireproof cabinets only accessible for authorized persons.

The destruction of an audit record can only be executed after signed authorization from the OWGTM auditor and the OWGTM Information Security Manager. A trace of the destroyed materials is kept for future references.

5.4.5 Audit log backup procedures

The audit logs are backed up using incremental and remote procedures.

5.4.6 Audit collection system (internal vs. external)

The collection systems for audit logs in OWGTM is a combination of automatic and manual processes, and is executed by the appropriate operating systems, software applications, and personnel operating these systems.

5.4.7 Notification to event-causing subject

No stipulations.

5.4.8 Vulnerability assessments

OWGTM executes regular vulnerability assessment by monitoring the activity logs, at reasonable frequencies. In depth assessments and checks are performed on a yearly basis, including conformance to disaster recovery plans. In the event that an assessment could not be performed or was delayed, the OWGTM will inform the involved parties and records of such an event and its cause will be kept for future reference.

This security analysis implies the identification of necessary tasks to correct detected vulnerabilities.

5.5 Records archival

This section includes the stipulations regarding record retention policies.

5.5.1 Types of records archived

The information and events archived are: - Information generated (at CA and RA) during the life cycle of all OWGTM certificates, - Contracts and agreements, - Audit logs stipulated in section 5.4 of this CPS.

5.5.2 Retention period for archive

Archived records and audit logs are kept. Records are retained for at least the validity of the involved certificates.

For the particular case of ECHONET certificates, the CA retains the records listed in section 5.5.1 for at least 7 years after the expiry or revocation of the corresponding certificate, consistent with section 6.3.1.

5.5.3 Protection of archive

Access to archived materials is restricted to authorized persons, and controls to ensure the archive integrity are enforced.

5.5.4 Archive backup procedures

Daily backup copies are executed. The main copy is kept in the principal OWGTM facility and stored inside a secured zone. Copies are periodically stored offsite.

5.5.5 Requirements for time-stamping of records

In addition to stipulations in 5.5.3, a time stamp is included in the digitally signed records. The time stamp needs not be of cryptographic nature.

5.5.6 Archive collection system (internal or external)

Archive collection is an internal task in the OWGTM that cannot be outsourced to third parties.

The only exception are authorized Registration Authority points, which are allowed to archive information collected during the certificate life-cycle. In such case, this information must be kept securely, accessible only for authorized persons, and made available to any internal or external auditing entity mandated by OWGTM.

5.5.7 Procedures to obtain and verify archive information

Only authorized personnel obtain access to the physical media containing archives, backups and other recorded information.

Integrity checks are performed automatically if the archive includes a digital signature.

5.6 Key changeover

OWGTM requires the creation of new keys for a CA needing to renew its certificate. Only in exceptional cases it can be accepted to repeat a CA Creation Ceremony maintaining the same keys created in a Hardware Security Module for a previous ceremony, in order to amend any error in the process.

When creating a new certificate for an entity, the validity period applied to this certificate will be constrained to the validity of the keys of the Certification Authority issuing it.

5.7 Compromise and disaster recovery

In the event that OWGTM systems and services are not available for a period greater than 12 hours, the Continuity Plan will be activated. This Continuity Plan seeks to ensure that the critical services (as stated in section 5.7.4) are available in less than 72 hours after the plan is activated.

The following sections summarize specific situations and the stipulated reaction in OWGTM. The detailed Continuity Plan is a confidential document.

5.7.1 Incident and compromise handling procedures

The Certification and/or Registration Authorities operating under the OWGTM are required to enforce the necessary controls to ensure and demonstrate that the Incident and Compromise Handling Procedures are effective. Involved people must be conveniently trained in their roles and responsibilities in the execution of their duties.

The following subsections disclose the procedures executed in such these events.

5.7.2 Computing resources, software, and/or data are corrupted

If the hardware or software resources are altered or suspected to have been altered, the OWGTM will stop normal operations until a secure environment is established. In parallel, an audit will be conducted in order to identify the cause and stipulate the necessary actions to avoid future iterations.

In the event digital certificates are issued during the uncertainty period and a risk exists that these certificates could be compromised, then those certificates will be revoked and subscribers will be notified of the need to reissue their certificates.

5.7.3 Entity private key compromise procedures

In the case a private key is compromised in the OWGTM architecture and in addition to stipulations in section 5.7.2, the subordinated entities depending on the compromised private key will be notified of this event and the necessary actions will be undertaken.

All certificates issued by entities subordinated to the compromised key from the time of the key's compromise and the certificate's revocation will be revoked, and the involved parties notified as stipulated in this CPS. Additional steps to re-issue the necessary certificates will be taken.

5.7.4 Business continuity capabilities after a disaster

In the event of a disaster (independently of its nature) that affects OWGTM's main facilities, and any services that are provided from these, the OWGTM Service Continuity Plan will be activated, ensuring that the services identified as "Critical" are available in less than 72 hours after the Plan activation. The rest of services would be available in the reasonable terms, as judged adequate for their importance and criticality level.

5.8 CA or RA termination

The causes that could imply the termination of a Certification or Registration Authority operating under the OWGTM are: - Private Key Compromise - A political or judicial decision - A Contract Termination after a breach of the corresponding Terms and Conditions

In the case a Certification Authority under OWGTM is forced to terminate its activities, the minimum actions to be executed are: - Immediately after there's a Termination decision, notify all certificate subscribers - Revoke all certificates under the CA. - Inform all relying parties that have a registered direct relationship with that Certification Authority about the termination of the certificate service provision. This will also terminate the accreditation granted to the Certification Authority to operate under OWGTM. - Publish a public notice of the termination within the repository section of the affected CA's web site, and undertake other public communications as deemed necessary to inform the wider relying party community.

In the case an OWGTM Root Certification Authority is terminated, this will imply the termination of the entire hierarchy dependent of that Root CA.

6. TECHNICAL SECURITY CONTROLS

This section describes the measures taken by Certification Authorities operating under the OWGTM. The OWGTM believes these controls are fundamental to provide trust to subscribers and all relying parties, and has therefore established the necessary means to ensure and demonstrate that these controls are enforced. These controls are under surveillance and audited both internally and externally by accredited bodies. The public manifests of these audits are published on a regular basis in the web site (<http://www.oiste.org/repository>).

6.1 Key pair generation and installation

Under the OWGTM, Key Pairs are generated under the necessary security levels and always occurring in secure physical facilities and under the adequate personnel control.

6.1.1 Key pair generation

Key Pairs of Certification Authorities operating in the OWGTM are generated and installed under a procedure compliant with applicable regulations. Main details of this procedure are: - The Root Certification Authority key creation ceremony is audited by an external qualified auditor. - Subordinated Certification Authorities are generated under direct supervision of internal auditors from WISeKey. - CA Ceremonies are executed by designated trusted personnel. - There's a pre-defined execution script that must be followed during the Ceremony. - During the Ceremony, enough audit track is recorded in order to proof that the Ceremony was executed as planned and without any security risk. - After the Ceremony, a Ceremony Report is generated and properly archived for future reference. Key pairs for the Root Certification Authorities in the OWGTM are generated in hardware security modules (HSM) accredited under the standards specified in section 6.2.1.

Key pairs for the Policy and Issuing Certification Authorities in the OWGTM may be generated in hardware security modules (HSM) accredited under the standards specified in section 6.2.1.

Key pairs for Subordinate Certification Authorities in the OWGTM may be generated in escrowable form, protected under dual control and split-knowledge at all times, and imported and operated within hardware security modules (HSM) under the standards specified in section 6.2.1.

For Subscriber Certificates, unless otherwise noted in this CPS, Subscriber is solely responsible for the generation of the Key Pair appropriate to the Certificate type being applied for.

6.1.2 Private key delivery to subscriber

Private keys corresponding to CA certificates are generated inside the CA cryptographic module and are never delivered to any subscriber.

If the specific subscriber certificate type allows the generation of the private key by the Registration Authority, the usage of password-protected encrypted software files, or smart-cards or other valid crypto-tokens is accepted.

6.1.3 Public key delivery to certificate issuer

Public keys of CA certificates are delivered to relying parties within the CA certificate itself, as described in section 6.1.4.

Public keys generated by, or for, the end-entities are sent to the Certification Authority through secure channels using the OWGTM Registration Authorities, as part of a certificate request in acceptable formats, such as PKCS#10 or other standard CSR format.

6.1.4 CA public key delivery to relying parties

The public keys of all Certification Authorities operating under the OWGTM Trust Model are included in the corresponding certificate and published and can be freely downloaded from its repository which is located at <http://www.oiste.org/repository>.

The ECHONET Root CA certificate is not distributed through browser or operating-system trust stores. Relying parties provision it as a trust anchor either from the OWGTM repository identified in section 2.1, or as embedded in the device or application by its manufacturer or integrator. The integrity of a trust anchor obtained from the repository must be verified against the fingerprint and Subject Key Identifier published in Appendix B.

6.1.5 Key sizes

Certificates issued under this CP/CPS use elliptic-curve keys: NIST P-256 or P-384 for ECHONET Subordinate CAs and ECHONET Device Certificates. The Root CA identified in Appendix B holds a NIST P-256 key.

The only hashing algorithms permitted are SHA-256 and SHA-384. SHA-1 SHALL NOT be used in any certificate or CRL issued under the ECHONET hierarchy.

6.1.6 Public key parameters generation and quality checking

The algorithm used in the ECHONET hierarchy for key generation is ECDSA over the NIST P-256 and P-384 curves. Key pairs are generated by a validated random source within the cryptographic module, and the CA rejects any public key that fails the quality checks of section 4.2.1.

6.1.7 Key usage purposes (as per X.509 v3 key usage field)

Key usage purposes for CA certificates is restricted to digital signature, CRL signature and certificate signing.

All subscriber certificates issued in the OWGTM contain the “KEY USAGE” and “EXTENDED KEY USAGE” attributes, as defined by the X.509v3 standard. More information is available in section 7 of this document.

6.2 Private Key Protection and Cryptographic Module Engineering Controls

The OWGTM has established controls to ensure that the risks derived from a private key compromise are managed and kept under reasonable levels. These controls are different for the main components (Certification Authorities) and end subscriber keys.

6.2.1 Cryptographic module standards and controls

Certification Authorities in the OWGTM are required to use Hardware Security Modules certified at least to FIPS 140-2 (or FIPS 140-3) Level 3 for CA key protection, and at least Level 2 for other PKI components. Common Criteria certification against a protection profile of equivalent assurance is also accepted.

6.2.2 Private key (n out of m) multi-person control

Private keys for Certification Authorities are always under multi-person control. Activation data needed to enable a Certification Authority will be shared in such a way that at least two authorized persons are needed to perform any sensitive operation on a Certification Authority, except where unattended operational restart of Issuing CAs is enabled.

Private keys for end-entities are under the sole control of the subscriber or authorized representative.

6.2.3 Private key escrow

Not applicable. Private key escrow is not provided for any certificate in scope of this CP/CPS, as stated in section 4.12.

6.2.4 Private key backup

Backup copies of CA private keys for all Certification Authorities under the OWGTM Trust Model are kept for routine recovery and disaster recovery purposes. Such keys are always stored in encrypted form within hardware cryptographic modules and associated key storage devices. Cryptographic modules used for CA private key storage meet the requirements of this CPS.

Private key backup for end-user subscribers, if supported for a certain certificate type, it would be implemented as described in section 4.12.

6.2.5 Private key archival

The CA shall not provide key archival services to subscribers. Private keys of expired or revoked CAs may be archived by being left in HSM or their backup.

6.2.6 Private key transfer into or from a cryptographic module

For Certification Authorities operating under the OWGTM Trust Model it is mandatory that key pairs are operated in Hardware Security Modules as defined in section 6.2.1. Private Keys can be transferred to adequate hardware security modules for back-up and recovery operations.

There's no stipulation for Keys belonging to other PKI participants.

6.2.7 Private key storage on cryptographic module

CA or RA private keys held on hardware cryptographic modules are stored in an encrypted form supported by the HSM vendor.

Private keys of ECHONET Device Certificates are generated and held in a hardware-backed key store (secure element, secure enclave or TPM) that prevents export of the private key in plaintext. Where a device cannot provide such a key store, the key must be held in an encrypted container and this limitation must be recorded by the Manufacturer and accepted by the PAA.

6.2.8 Method of activating private key

The private key in Certification Authorities in the OWGTM is activated by initiating the PKI Software and activating the HSM where the key is stored. This process requires at least a dual-person control, except for Issuing CAs where automatic key activation in case of system failure or restart is allowed.

The activation of Subscriber's private key is stipulated in section 6.4.

6.2.9 Method of deactivating private key

The private key in Certification Authorities is deactivated by shutting-down the associated server or by terminating the PKI software or by extracting or shutting-down the HSM that contains the key. This task can be done by a System Administrator and, when planned, has to be notified and authorized to/from the CA Responsible.

Deactivating RA or other end-user private keys based in hardware is performed by the extraction of the secure device (smart-card or other accepted crypto-tokens) from the workstation it is used.

Deactivating of other end-user subscriber private keys, while not based in hardware, is accomplished by shutting down the device where the private key is stored. The subscriber must take all reasonable measures to avoid unauthorized use of the device.

6.2.10 Method of destroying private key

The procedure to destroy a private key is initiated in the following cases: - Private Key is no longer used and it's mandated its destruction - The token or HSM containing the key has deteriorated to an extent that prevents normal usage - A lost or stolen token is found, and the keys it contained are suspected to be compromised

A private key can be destroyed by the key owner or a legal representative. In such cases the corresponding certificate will be revoked, and the community will be notified. The procedure used to destroy the private key depends on the particular container holding it, being responsibility of the individual executing the destruction doing it in an appropriate way. In particular, for private keys associated to CAs, this task must be executed under dual control and appropriate tracking information must be recorded.

6.2.11 Cryptographic Module Rating

No stipulation additional to section 6.2.1.

6.3 Other aspects of key pair management

This section includes additional stipulations regarding key pair management.

6.3.1 Public key archival

Public keys in the OWGTM trust model are archived for a period of 7 years after the expiry or revocation of the corresponding digital certificate.

6.3.2 Certificate operational periods and key pair usage periods

The fully operational period for a certificate starts at the issuance and ends with the expiration or revocation of the certificate. Certificates in the ECHONET hierarchy have no scheduled expiration, unless a fixed validity has been requested for a product line as stated below, so for these certificates the operational period ends only on revocation.

The validity period for key pairs is stipulated in the following table:

Certificate Type	Maximum Validity Period
Root CA (identified in Appendix B)	No scheduled expiration. notAfter = 31 December 9999, 23:59:59 UTC (99991231235959Z), the RFC 5280 encoding for a certificate with no well-defined expiry (section 4.1.2.5)
ECHONET Subordinate CA	No scheduled expiration. notAfter = 99991231235959Z
ECHONET Device Certificate	No scheduled expiration. notAfter = 99991231235959Z

ECHONET Device Certificates are issued without a scheduled expiry because the appliances they identify (heat-pump water heaters, storage batteries, EV chargers and similar equipment) remain in service for fifteen years or more, have no reliable field mechanism for certificate renewal, and must be able to authenticate for as long as they operate. A Manufacturer may request a fixed **notAfter** value for a given product line, in which case that value applies to every Device Certificate issued for it.

ECHONET Subordinate CA certificates are issued without a scheduled expiry, on the same basis as the Root CA, so that no Subordinate CA truncates the validity of the Device Certificates it issues.

It must be understood that the validity period of a certificate can be limited by the validity of the issuing Certification Authority. In the ECHONET hierarchy neither the Root CA nor the Subordinate CAs impose such a limit.

Because no certificate in the hierarchy expires, a key cannot be retired by letting its certificate lapse, and the following consequences apply:

- **Revocation is the only means of ending the validity of a certificate.** A revoked certificate remains listed on the CRL of its issuer permanently (section 7.2.2).
- **A Subordinate CA is retired without being revoked.** Revoking a Subordinate CA certificate would invalidate every Device Certificate beneath it. A Subordinate CA whose key is to be retired is therefore subject to the key changeover procedure of section 5.6, after which it ceases issuing Device Certificates. Its certificate remains valid, and it continues to issue CRLs under section 4.9.7, for as long as Device Certificates issued beneath it remain in service. It is revoked only on the grounds of section 4.9.1.2.
- **Retention periods keyed to expiry run from revocation.** The periods of sections 5.5.2 and 6.3.1 start on revocation for a certificate with no scheduled expiration, so its records and public key are retained for as long as it has not been revoked, and for the stated period thereafter.

The certificates are operational for signature validation and decryption from the issuance to the end of the archival period stated in 6.3.1.

The operational period of subscriber certificates can be restricted by the ECHONET Requirements listed in Appendix D.

6.4 Activation data

This section stipulates the management of the data necessary to activate the private keys.

6.4.1 Activation data generation and installation

Activation data for Certification Authorities are generated and stored in cryptographic tokens and/or smart cards and are only used by authorized persons. In addition, these tokens require a password or PIN in order to enable the activation process.

Activations requiring a multi-person control will be enforced by splitting the activation data in several tokens.

For ECHONET Device Certificates, the private key is generated inside the device secure element or equivalent hardware key store, or injected in a controlled manufacturing environment. In either case: - The key store must enforce access control such that the private key can only be used by the device firmware for the purposes stated in section 1.4.1. - Where keys are injected in a factory, the injection environment must be physically and logically controlled by the Manufacturer, key material must be transported encrypted, and any copy held outside the device must be destroyed once injection is confirmed. - No shared or default activation secret may be used across devices.

Private Keys for Registration Authority operator certificates, if protected in a hardware device, require the usage of a password or PIN code of eight or more characters in order to activate the device where the key is stored.

6.4.2 Activation data protection

Only the authorized persons know the password or PIN to activate the private keys. In the case of end-entities, only the certificate subscriber is entitled to know this information.

In all cases, the owner of the activation data is required to safeguard the secrecy of this information.

6.4.3 Other aspects of activation data

No stipulation.

6.5 Computer security controls

The details of this information are classified and therefore not made public. The documents describing Computer Security Controls are only available for the people involved in the OWGTM and only disclosed to accredited external parties for auditing purposes.

Certification and Registration Authorities operating under the OWGTM Trust Model are required to meet these Security Controls. The compliance is periodically enforced by an auditing procedure.

6.5.1 Specific computer security technical requirements

OWGTM enforces the use of the appropriate procedures and technical measures and systems in order to effectively control security risks. These include, but not limited to: - Strong password policies - Constant improvement of administration and operating procedures - Physical isolation of confidential systems - Antivirus and anti-malware detection systems - Periodic internal security reviews

6.5.2 Computer security rating

OWGTM establishes the computer ratings to be met by the Certifications and Registration Authorities operating under the Trust Model. Compliance with these ratings is ensured by periodic internal audits.

6.6 Life cycle technical controls

This information is classified and is therefore not disclosed in detail. The detailed documents are available for review by external auditors after the appropriate authorization process.

6.6.1 System development controls

Systems are developed using the WISeKey KeySteps Methodology, which ensures the security and quality by setting a series of policies and operational and technical procedures controlling the building of the PKI components during all the phases of the project.

Authenticity and integrity of critical software components must be checked before they are enabled in a production environment, by using code signing or other acceptable methods.

6.6.2 Security management controls

The OWGTM recommends following the ISO27000 security management approach. In particular WISeKey, as main operator of the Trust Model follows an informal adoption of such security standards.

6.6.3 Life cycle security controls

Life cycle and change-related security controls are ensured by the WISeKey KeySteps Methodology.

6.7 Network security controls

The OWGTM enforces the adoption of effective controls to minimize any risk related to Network Security. The detailed information about these controls is classified and only made available for external auditors after the appropriate authorization process.

In particular, the server used for the OWGTM Root CA are off-line systems, physically disconnected from any computer network, and all communication of sensitive information is protected using encryption and digital signature techniques.

6.8 Time-stamping

No stipulation.

7. CERTIFICATE AND CRL PROFILES

All certificates issued under the OWGTM are compliant to: - ITU-T Recommendation X.509 (1997): Information Technology - Open Systems Interconnection - The Directory: Authentication Framework, June 1997 - RFC 5280: Internet X.509 Public Key Infrastructure Certificate and CRL Profile, April 2002 ("RFC 5280").

7.1 Certificate profile

The OWGTM defines different certificate profiles corresponding to the allowed certificate types issued under the different hierarchies.

The general certificate profiles are: - ECHONET Root CA Certificate: self-signed trust anchor of the ECHONET hierarchy. - ECHONET Subordinate CA Certificate: issues ECHONET Device Certificates, technically constrained as stated in section 7.1.5. - ECHONET Device Certificate: identifies an individual ECHONET device and is used for client authentication and key agreement.

The different profiles are mainly differentiated by the appropriate combination of values in the "Key Usage", "Extended Key Usage" and/or the use of particular Policy Identifiers.

All three profiles carry a **notAfter** value of 99991231235959Z, encoded as GeneralizedTime as required by RFC 5280 section 4.1.2.5, unless a fixed validity has been requested for a product line under section 6.3.2.

The OWGTM must ensure that the certificate profiles are aligned with this section and with any applicable ECHONET Requirement.

7.1.1 Version number(s)

All certificates in the OWGTM conform to X.509 Version 3.

7.1.2 Certificate extensions

The following extensions are used in the ECHONET hierarchy. Any extension not listed is not included.

Root CA Certificate (existing certificate, identified in Appendix B)

Extension	Critical	Value
basicConstraints	Yes	cA = TRUE, no pathLenConstraint
keyUsage	Yes	digitalSignature, keyCertSign, cRLSign
subjectKeyIdentifier	No	C5:3B:1F:4C:A7:B9:26:D1:FD:B4:85:29:3F:...
authorityKeyIdentifier	No	Same as the subjectKeyIdentifier, the certificate being self-signed

The Root CA certificate was issued on 12 August 2024 and pre-dates this CP/CPS. This profile is descriptive of that certificate and is not a specification for a future issuance.

ECHONET Subordinate CA Certificate

Extension	Critical	Value
basicConstraints	Yes	cA = TRUE, pathLenConstraint = 0
keyUsage	Yes	keyCertSign, cRLSign
certificatePolicies	No	2.16.756.5.14.8.3.2
subjectKeyIdentifier	No	Hash of the public key
authorityKeyIdentifier	No	Key identifier of the ECHONET Root CA
cRLDistributionPoints	No	HTTP URI of the Root CA CRL
authorityInfoAccess	No	HTTP URI of the Root CA certificate
nameConstraints	Yes	As stated in section 7.1.5, for manufacturer-operated Subordinate CAs

ECHONET Device Certificate

Extension	Critical	Value
basicConstraints	Yes	cA = FALSE
keyUsage	Yes	digitalSignature, keyAgreement
extKeyUsage	No	id-kp-clientAuth (1.3.6.1.5.5.7.3.2)
certificatePolicies	No	2.16.756.5.14.8.3.3
subjectKeyIdentifier	No	Hash of the public key
authorityKeyIdentifier	No	Key identifier of the issuing ECHONET Subordinate CA
cRLDistributionPoints	No	HTTP URI of the Subordinate CA CRL
authorityInfoAccess	No	HTTP URI of the Subordinate CA certificate

ECHONET Device Certificates do not contain a subjectAltName extension, a dNSName, an iPAddress or an rfc822Name.

7.1.3 Algorithm object identifiers

The ECHONET hierarchy uses ECDSA exclusively. The permitted signature algorithm identifiers are:

Certificate	Signature algorithm (applied by its issuer)	Subject public key
Root CA (existing, self-signed)	ecdsa-with-SHA256 (1.2.840.10045.4.3.2)	id-ecPublicKey on prime256v1 / secp256r1 (NIST P-256)
ECHONET Subordinate CA	ecdsa-with-SHA256 (1.2.840.10045.4.3.2), being the algorithm of the signing Root CA	id-ecPublicKey on secp256r1 or secp384r1
ECHONET Device Certificate	ecdsa-with-SHA256 (1.2.840.10045.4.3.2), or ecdsa-with-SHA384 (1.2.840.10045.4.3.3) where the issuing Subordinate CA holds a P-384 key	id-ecPublicKey on secp256r1 or secp384r1

The Root CA holds a P-256 key, so the effective security level of every chain built to this trust anchor is that of P-256, irrespective of the curve used lower in the hierarchy.

No certificate issued beneath the Root CA under this CP/CPS uses RSA keys or the SHA-1 hash algorithm. The Root CA certificate itself pre-dates this CP/CPS and its algorithms are recorded in Appendix B.

7.1.4 Name forms

For CA certificates, the Subject Name combines appropriate values of commonName, organizationName and countryName to form an identifier that uniquely identifies the CA and distinguishes it from other CAs in the Trust Model.

For ECHONET Device Certificates, the Subject Name is composed as follows:

Attribute	Presence	Content
countryName (C)	Required	ISO 3166-1 alpha-2 code of the Manufacturer's jurisdiction of incorporation
organizationName (O)	Required	Legal name of the Manufacturer, as verified under section 3.2.2
organizationalUnitName (OU)	Required	MFRCODE: followed by the three-byte Manufacturer Code (EPC 0x8A), in uppercase hexadecimal, octets separated by colons
organizationalUnitName (OU)	Required	PRODCODE: followed by the Product Code (EPC 0x8C) assigned by the Manufacturer
commonName (CN)	Required	The Production Number (EPC 0x8D) of the individual device, unique within the Product Code

The combination of Manufacturer Code, Product Code and Production Number is unique across the ECHONET hierarchy and satisfies the uniqueness requirement of section 3.1.5.

7.1.5 Name constraints

The Root CA identified in Appendix B does not assert name constraints.

An ECHONET Subordinate CA operated by a Manufacturer SHALL be technically constrained by a critical nameConstraints extension whose permittedSubtrees limit the directoryName to the Manufacturer's own organizationName and countryName, and whose excludedSubtrees exclude dNSName, iPAddress and rfc822Name in their entirety. The Subordinate CA SHALL additionally enforce, by configuration, that it issues only for the Manufacturer Codes assigned to that Manufacturer.

Subordinate CAs operated by WISeKey may be constrained by the same means where a Manufacturer requests it.

7.1.6 Certificate policy object identifier

An object identifier (OID) is a unique number that identifies an object or policy. The OIDs are administered by the OWGTM and listed in the Appendix C, "OID Inventory".

7.1.7 Usage of Policy Constraints extension

No stipulation.

7.1.8 Policy qualifiers syntax and semantics

Unless disallowed by the applicable requirements, certificates may contain information in the Certificate Policy extension.

7.1.9 Processing semantics for the critical Certificate Policies extension

The "Certificate Policy" extension identifies the Policy that the OWGTM assigned explicitly with a certificate policy. Software Applications requiring a specific certificate profile to process a digital signature must check this extension in order to verify the suitability of the certificate for the intended purpose.

7.2 CRL profile

In general, CRLs generated under the OWGTM Trust Model are compliant with RFC 5280 (Internet X.509 Public Key Infrastructure Certificate and CRL Profile, April 2002).

7.2.1 Version number(s)

CRLs conforming to X.509 Version 2 are supported in the OWGTM.

7.2.2 CRL and CRL entry extensions

CRL must include the following minimum extensions, as defined by the above standard: - CRL Number - Authority Key Identifier - Revocation date - Reason code

The permitted values of the "Reason Code" are those of RFC 5280, and their use is appropriately communicated in the Subscriber Agreement.

In particular, the use of the reason "keyCompromise", when the revocation is done by the CA or RA, is regulated as described in section 4.9.12.

An entry for a revoked certificate with no scheduled expiration is never removed from the CRL, since such a certificate never reaches the expiry after which RFC 5280 permits its removal.

7.3 OCSP profile

Not applicable. No OCSP responder is operated for the ECHONET hierarchy; revocation status is published exclusively by CRL, as stated in section 7.2.

8. COMPLIANCE AUDIT AND OTHER ASSESSMENTS

OWGTM monitors and ensures compliance to legal, security and industry requirements, in all levels of the Trust Model, through internal and external audits.

8.1 Frequency or circumstances of assessment

Every Certification Authority operating under a Root regulated by this CP/CPS, and every dependent Registration Authority, SHALL provide the PAA with a self-assessment against this CP/CPS: - **Initially**, before the CA is enabled for production issuance. No Subordinate CA Certificate is issued, and no Registration Authority is accredited, until the initial self-assessment has been submitted to and accepted by the PAA. - **Annually** thereafter, within twelve months of the previous self-assessment.

The OWGTM, for its part, is bound to audit any participant seeking to operate under its Roots. The PAA will conduct, or commission, an assessment of any such participant before admission and whenever it considers it necessary thereafter, and the participant is contractually obliged to submit to that assessment and to grant the access required to perform it.

The detailed internal audit scheme — the assessment criteria, the reporting format and the qualification of the assessors — will be defined by the PAA and published in a future revision of this document. Until it is published, self-assessments are performed against the provisions of this CP/CPS as a whole.

8.2 Identity/qualifications of assessor

The assessor will be selected when an audit or assessment is required. Any company or professional whose services are contracted as auditor or assessor will be required to fulfil these requirements: - Adequate and demonstrable capability and experience in PKI audit and information security assessment. - Independence from the operational management of the entity being assessed. The assessor reports to the PAA and not to the audited CA or RA.

A self-assessment under section 8.1 is prepared by the assessed entity itself, and must be signed by a person empowered to bind that entity.

The ECHONET hierarchy is not assessed under WebTrust or any equivalent browser root-program audit scheme. Assessment is performed against this CP/CPS by the PAA, or by an assessor it appoints, and the resulting Audit Statement Report is published as stated in section 8.6. The qualification criteria for appointed assessors will be set out in the internal audit scheme announced in section 8.1.

8.3 Assessor's relationship to assessed entity

The OWGTM audit policy does not allow any kind of legal, organizational or other relationship with the external auditor that would result in a conflict of interests.

8.4 Topics covered by assessment

The OWGTM establishes the need for audit and accreditation of the following: - The Root CA and the Subordinate CAs owned or operated by WISeKey. These services are assessed as stated in sections 8.1 and 8.2. - Subordinate CAs owned and/or operated by Manufacturers or other third parties, and Registration Authorities. These services must meet the practices stipulated in this CP/CPS and the certificate policies they are entitled to issue under. A manufacturer-operated Subordinate CA is admitted only after the initial self-assessment of section 8.1 has been accepted and the OWGTM has completed its own accreditation assessment of the participant.

A participant that fails to submit a self-assessment when due, or that refuses or obstructs an assessment conducted or commissioned by the PAA, is in breach of this CP/CPS. The PAA may suspend its issuance rights under section 8.5 and, where the breach is not remedied, the Subordinate CA Certificate is revoked under section 4.9.1.2.

8.5 Actions taken as a result of deficiency

In the case a deficiency is identified, the OWGTM will adopt and will be responsible for all necessary corrective measures.

In the case of a severe deficiency affecting the reliable operation of a Certification or a Registration Authority, the OWGTM could decide to temporarily suspend the activities of the affected systems or services until the deficiency is solved.

8.6 Communication of results

All assessment results will be conformed as: - Detailed Report. This document includes all the topics covered by the executed assessment program in detail. The detailed report is deemed private and only available to the following parties: - Certification Authority owner - OWGTM Policy Approval Authority - The ECHONET Consortium, upon reasoned request - Audit Statement Report. This document only includes a formal statement from the auditor and reflects the result of the assessment, listing the topics covered and a global result. The summarized report is deemed public and is only published in the OWGTM and Issuing Repository.

9. OTHER BUSINESS AND LEGAL MATTERS

This section includes the stipulations for business and legal matters and should be understood as having a contractual value by all the PKI participants.

9.1 Fees

The fees applicable to the Certification Services covered by this CPS can be subject to variation according to specific agreement with the participants in the service. The detailed information of the fees is made available for the subscribers or other affected parties before enabling such services.

9.1.1 Certificate issuance or renewal fees

The issuance of certificates in the OWGTM is considered a commercial service and therefore subject to fees. The fees depend on the certificate and project and are agreed before making it available to subscribers.

9.1.2 Certificate access fees

OWGTM doesn't enforce stipulations for certificate access fees. In general, any participant shouldn't apply fees on the access to certificate information made public in the different repositories.

9.1.3 Revocation or status information access fees

OWGTM doesn't enforce stipulations for revocation or status information access fees. In general, the Issuing CA shouldn't apply fees on the access to certificate information made public in the different repositories.

9.1.4 Fees for other services

The operators of Issuing CAs in the OWGTM can set fees for different commercial services provided to parties willing to participate in the Trust Model. This includes, but not limited to: - Managed PKI Services - CA Signing Services - CA Hosting and operation services

9.1.5 Refund policy

The refund policy applicable to commercial services provided by WISEKey is included in the "Subscriber agreement" and/or general Terms and Conditions communicated to the end-user when providing the service. Other refund policies can be established and, in such cases, must be effectively communicated to all affected parties.

9.2 Financial responsibility

The OWGTM established the adequate controls to ensure that the different levels of financial responsibility are met by the different participants, according to their impact in the trust model.

9.2.1 Insurance coverage

For the Root CA, Issuing CAs and the certification services provided directly by WISEKey, it is maintained an Errors and Omissions insurance policy that covers the liability expressed in section 9.8. For affiliates and corporate customers acting as Certification or Registration Authorities, the contractual terms agreed among the parties ensure the assumed responsibilities for each party and transfer the requirement for appropriate insurance for the transferred liabilities.

9.2.2 Other assets

No stipulation.

9.2.3 Insurance or warranty coverage for end-entities

The maximum liability per subscriber certificate issued under the OWGTM is to be established in the applicable Subscriber Agreement published by the Issuing CA.

9.3 Confidentiality of business information

In general, an Issuing CA under the OWGTM may not disclose the confidential information of a subscriber, or use that information for any purpose, except: - To its staff requiring the information for the purposes of this CPS or for delivery of the services. - With the explicit consent of the subscriber. - If required to do so by any law, or an applicable agreement.

9.3.1 Scope of confidential information

Information released to subscriber(s) or relying parties by Issuing CA may be considered confidential.

All Issuing CA under the OWGTM shall keep the following types of information confidential and maintains reasonable controls to prevent the exposure of such records to non-trusted personnel. - All private keys - Any activation data used to access private keys or gain access to the CA system - Any business continuity, incident response, contingency, and disaster recovery plans - Any other security practices, measures, mechanisms, plans, or procedures used to protect the confidentiality, integrity or availability of information - Any information held by the Issuing CA in accordance with Section 9.4 - Any transactional, audit log and archive record identified in Section 5.4 or 5.5, including certificate application records and documentation submitted in support of certificate applications whether successful or rejected. - Transaction records, financial audit records and external or internal audit trail records and any audit reports (with the exception of an auditor's letter confirming the effectiveness of the controls set forth in this CPS) - All information classified explicitly as "PRIVATE", "CONFIDENTIAL" or "STRICTLY CONFIDENTIAL" when generated or exchanged among involved parties.

9.3.2 Information not within the scope of confidential information

The following information shall be deemed as non-confidential: - All information contained in the issued certificates and Certificate Revocation Lists (CRLs) including all information that can be derived from such. - All information classified expressly as "PUBLIC".

9.3.3 Responsibility to protect confidential information

The OWGTM Issuing CAs are responsible of the protection of the confidential information generated or communicated during all operations. Delegated parties, as the entities managing subordinate Issuing CAs or Registration Authorities, are responsible for protecting confidential information that has been generated or stored by their own means.

For end entities, the certificate subscribers are responsible to protect their own private key and all activation information (i.e. passwords or PIN) needed to access or use the private key.

9.4 Privacy of personal information

The Issuing CAs operating in the OWGTM must publish their own Privacy Policy and communicate it adequately to the certificate subscribers. This Policy must be compliant with the applicable requirements for international commercial services, and specifically with the ECHONET Requirements, the European General Data Protection Regulation (GDPR) and, for subscribers established in Japan, the Act on the Protection of Personal Information (APPI).

In general, it must be understood that the CAs act as a “Data Controller” and the RAs and other parties involved in certificate management are “Data Processors” or, in certain occasions, “Joint Controllers”.

9.4.1 Privacy plan

WiSeKey publishes the Privacy Policy and other related materials in <https://www.wisekey.com/repository>.

9.4.2 Information treated as private

Personal information about an individual that is not publicly available in the contents of a certificate or CRL is considered private.

9.4.3 Information not deemed private

For personal information the provisions of section 9.3.2 apply respectively.

9.4.4 Responsibility to protect private information

The OWGTM ensures the compliance of the legal obligations for Certification Authorities, Registration Authorities and other entities operating under the OWGTM Trust Model. Each of these participants is responsible to protect the private information that has been provided by subscribers or other participants in the issuance and maintenance of digital certificates.

9.4.5 Notice and consent to use private information

In order to perform the certification provisioning service, the Issuing CAs and other parties interacting with certificate subscribers are required to obtain the consent to use the subscriber’s personal information.

This consent is understood by the explicit acceptance of the “ECHONET Certificate Subscriber Agreement” by the Manufacturer during the certificate request process. This acceptance is recognized by the subscriber’s acceptance to obtain and install the certificate.

9.4.6 Disclosure pursuant to judicial or administrative process

The participants in the OWGTM will disclose personal information of the participants if required by a judicial or administrative process, upon presentation of appropriate orders in accordance with the Applicable Laws of the country where the Certification Authority operates.

9.4.7 Other information disclosure circumstances

No stipulation.

9.5 Intellectual property rights

All Intellectual Property rights, including the digital certificates and CRLs issued by the OWGTM Root CAs, Object Identifiers, this CPS and the different CP are owned by the OISTE Foundation.

The private and public keys are the property of their respective owners.

Any commercial or protected trademark included in the Distinguished Name of a certificate is under responsibility of the certificate subscriber.

9.6 Representations and warranties

This section includes general stipulations, specific terms can be stipulated in the appropriate Certificate Policy for a given certificate type and users community. If such is the case, specific Subscriber, Relying Party and other agreements will be distributed among the parties.

9.6.1 CA representations and warranties

OWGTM Root CAs will: - Establish a chain of trust by issuing a certificate, which is a self-signed certificate - Ensure that the Root signs any subordinate CAs issued under the OWGTM hierarchy - Properly conduct the verification process described in section 3.2 - Ensure the accuracy and completeness of any part of the certificate information which is generated or compiled by the OWGTM, according to the applicable Certification Policy - Ensure that all relevant information concerning a certificate is recorded (electronically or otherwise) for an appropriate period of time, and in particular, for the purpose of providing evidence for the purposes of legal proceedings - Utilize trustworthy systems, procedures and human resources in performing its services - Comply with any other relevant provisions of the relevant CP or CPS, and other approved documents.

All CAs in the OWGTM will: - Operate according to the requirements of this CPS and any applicable SLA. - Ensure at the time it issues a certificate, that the certificate contains all the elements required by the CP or PDS. - Manage their keys in accordance with Section 6.2 Private Key Protection and Cryptographic Module Engineering Controls. - Ensure the availability of a Certificate Directory and CRL - Promptly revoke a certificate if required. - In particular and where applicable, CAs will respect the warranties and obligations set by the ECHONET Requirements.

9.6.2 RA representations and warranties

The Registration Authorities operating under the OWGTM warrant that: - Will operate according to the requirements of this CPS. - Their Certificates meet all material requirements of this CPS. - No errors have been introduced in the Certificate information by the entities approving the Certificate Application as a result of a failure when managing the Certificate Application. - There are no material misrepresentations of fact in the Certificate at the entities approving the Certificate Application or issuing the Certificate. - Availability of revocation services (when applicable) and use of a repository conforming with the applicable CPS in all material aspects.

Registration Authority commercial contracts and agreements could include additional warranties.

9.6.3 Subscriber representations and warranties

The Subscribers of certificates issued under the OWGTM must warrant that: - All information supplied by the Subscriber and contained in the Certificate is true and valid. - All representations made by the Subscriber in the submitted Certificate Application are true and valid. - His or her private key is protected and that no unauthorized person has ever had access to the Subscriber's private key. - An obligation and warranty that it will not install and use the Certificate(s) until it has reviewed and verified the accuracy of the data in each Certificate. - An obligation and warranty to install the Certificate and its associated Private Key only in the device that the Certificate identifies, and to use the Certificate solely in compliance with all applicable laws, solely for authorized company business, and solely in accordance with the Subscriber Agreement. - The Certificate is being used exclusively for authorized and legal purposes, consistent with this CPS. - Each digital signature created using the private key corresponding to the public key listed in the Certificate is the digital signature of the Subscriber and the Certificate has been accepted and is operational (not expired or revoked) at the time the digital signature is created. - The Subscriber is an end-user Subscriber and not a CA, and is not using the private key corresponding to any public key listed in the Certificate for purposes of digitally signing any Certificate (or any other format of certified public key) or CRL, as a CA or otherwise. - An obligation and warranty to promptly cease using a Certificate and its associated Private Key, and promptly request that the Certification Authority revoke the Certificate, in the event that: (a) any information in the Certificate is or becomes incorrect or inaccurate, or (b) there is any actual or suspected misuse or compromise of the

Subscriber's Private Key associated with the Public Key listed in the Certificate. - An obligation and warranty to promptly cease all use of the Private Key corresponding to the Public Key listed in an Certificate upon expiration or revocation of that Certificate. The "Subscriber agreement" could include additional warranties.

9.6.4 Relying party representations and warranties

Before relying on a certificate or a digital signature, relying parties must: - Validate the certificate and digital signature (including by checking whether or not it has been revoked, expired or suspended) - Ascertain and comply with the purposes for which the certificate was issued and any other limitations on reliance or use of the certificate that are specified in this CPS.

If a relying party relies on a digital signature, or certificate, in circumstances where it has not been validated, it assumes all risks with regard to it (except those that would have arisen had the relying party validated the certificate), and is not entitled to any presumption that the digital signature is effective as the signature of the subscriber or that the certificate is valid.

Relying parties must also comply with any other relevant obligations specified in this CPS including those imposed on the entity when it is acting as a subscriber.

Additionally, the relying party should consider the certificate type. The final decision concerning whether or not to rely on a verified digital signature is exclusively that of the relying party.

The "Relying party agreement" could include additional warranties.

9.6.5 Representations and warranties of other participants

No stipulation.

9.7 Disclaimers of warranties

Other Disclaimer of warranties (if existing) is included as part of the agreement presented to each PKI participant, or included in other documents published by the Issuing CA.

9.8 Limitations of liability

Liability limitations are regulated in the contractual agreement between the concerned parties. If applicable such concepts are specified in the Subscriber, Relying Party or other commercial agreements made among the participants.

Subject to the foregoing limitations, OWGTM's aggregate liability limit towards all End users, Relying Parties and any other entities that are not Subordinate PKI Entities for the whole of the validity period of certificates issued by the Root CA (unless revoked or suspended prior to its expiry) towards all persons with regard to such certificates is CHF 5,000,000.00 (Five Million Swiss Francs), with a maximum aggregate per year liability on such certificates of CHF 500,000.00 (Five Hundred and Thousand Swiss Francs). The OISTE Foundation delegates in WISEKey, as lead operator, this liability, according to a formal agreement executed between the parties, and that WISEKey ensures via an appropriate "Errors and Omissions" insurance.

9.9 Indemnities

Indemnities are regulated in the contractual agreement between the concerned parties. If applicable such concepts are specified in the CPS published in the Subscriber, Relying Party or other commercial agreements made among the participants.

9.10 Term and termination

This section refers to the times and validity periods related to this document.

9.10.1 Term

This Document becomes effective once published in the OWGTM Repository.

9.10.2 Termination

This Document (at the current version) is valid until replaced by a new version.

9.10.3 Effect of termination and survival

The Certificates issued during the validity period of the version of this document are bound to the clauses hereby included until the expiration or revocation of these certificates.

The termination of the CP/CPS shall be without prejudice to the responsibility to protect confidential and personal information.

9.11 Individual notices and communications with participants

Notices to subscribers must be sent to the physical, postal, facsimile or email address of the subscriber, which is included in its registration information, or to another address that the subscriber has specified to the sender. Reasonable measures to ensure the reception of the notices are taken.

9.12 Amendments

The OWGTM can unilaterally amend this document, by attaining adhering to the following procedure: - The modification needs to be justified under legal and technical considerations. - Any modification in the CPS cannot contradict the stipulations in the related CP, and vice-versa. - There is a modification procedure and change management for these amendments. - Any implications to the participants due to such amendments will be conveniently notified.

9.12.1 Procedure for amendment

The entity with the authority to make and approve any change in the CPS and the related CP in the OWGTM is the Policy Approval Authority (PAA, described in section 1.5 of this document), which reviews the change request, assesses whether the change request is required, and approves the changes.

A change can only be made to the approved documents once approval has been granted by the PAA.

On the assumption that the PAA decides to modify the CPS or a particular CP, a new version of the document will be generated. The version of the document (exposed in all the pages of the document) is controlled with two numbers separated by a period. The first number (major version) is incremented if the new version could affect the acceptance of the certificates by the users. The second number (minor version) is incremented if the amendment is not considered to affect the certificate acceptance criteria. The version numbers are not encoded in the OID identifying the document, which is stable across versions as stated in Appendix C.

Once a new version of the document is approved, the procedures stipulated in section 9.12.2 will be executed.

9.12.2 Notification mechanism and period

Any modification in this document will be published in the OWGTM website (<http://www.oiste.org/repository>) and affected participants will be directly notified if necessary.

In particular, it is not considered necessary to directly notify participants of “minor version” changes of the documents.

In the case of a change in the “major version” of a document, the OWGTM may notify the affected participants with a digitally signed electronic message.

9.12.3 Circumstances under which OID must be changed

The OID identifying this CP/CPS does not change between versions of the document.

The OID identifying a Certificate Policy is changed only where the change in policy is such that certificates already issued under the previous OID could no longer be considered to have been issued under the new one. In that case a new OID is allocated from the arc in Appendix C and the previous OID is retired, not reused.

9.13 Dispute resolution provisions

As agreed between the parties by the acceptance of Subscriber and/or Relying Party agreements. If no prior agreement was made to the dispute resolution mechanism, general rules of law shall apply.

9.14 Governing law

The CP, the CPS and the operations of the OWGTM are all governed by the laws of Geneva, Switzerland.

9.15 Compliance with applicable law

All related parties shall comply with all applicable Swiss laws, rules, regulations, ordinances, and directives, and all provisions required thereby to be included in this CPS are hereby incorporated herein by reference.

Applicable national laws can affect parties operating Certification Authorities in different jurisdictions.

9.16 Miscellaneous provisions

This section includes miscellaneous contractual and legal clauses.

9.16.1 Entire agreement

All provisions made in this CPs and the associated CP apply to all Certification and Registration Authorities operating under the OWGTM and its subscribers.

Agreements or supplementary agreements by word of mouth are not allowed.

9.16.2 Assignment

Parties to this CPS may not assign any of their rights or obligations under this CPS or applicable agreements without the written consent of WISeKey.

9.16.3 Severability

Should individual provisions of this CPS prove to be ineffective or incomplete, this shall be without prejudice to the effectiveness of all other provisions.

The ineffective provision will be replaced by an effective provision deemed as most closely reflecting the sense and purpose of the ineffective provision. In the case of incomplete provisions, amendment will be agreed as deemed to correspond to what would have reasonably been agreed upon in line with the sense and purposes of this CPS, had the matter been considered beforehand.

9.16.4 Enforcement (attorneys' fees and waiver of rights)

No stipulation.

9.16.5 Force Majeure

Force Majeure clauses, if existing, are included in the "Subscriber Agreement".

9.17 Other provisions

No stipulation.

Appendix A: Glossary

Acronyms

Acronym	Description
APPI	Act on the Protection of Personal Information (Japan)
CA	Certificate Authority or Certification Authority
CP	Certificate Policy
CPS	Certification Practice Statement
CRL	Certificate Revocation List
CSR	Certificate Signing Request
ECDSA	Elliptic Curve Digital Signature Algorithm
ECHONET	The ECHONET Consortium, and the ECHONET Lite family of specifications it publishes
EPC	ECHONET Property Code
FIPS	(US Government) Federal Information Processing Standard
GDPR	General Data Protection Regulation (European Union)
HSM	Hardware Security Module
HTTP	Hypertext Transfer Protocol
IETF	Internet Engineering Task Force
ITU	International Telecommunication Union
NIST	National Institute of Standards and Technology
OID	Object Identifier
PAA	Policy Approval Authority
PKI	Public Key Infrastructure
RA	Registration Authority
RFC	Request for Comments (at IETF.org)
SHA	Secure Hashing Algorithm
TLS	Transport Layer Security
X.509	The ITU-T standard for Certificates and their corresponding authentication framework

Definitions

Definition	Description
Applicant	An entity applying for a Certificate.
Certificate	An electronic document, conformant to X.509v3, digitally signed by a Certificate Authority, that binds a Public Key to an identity.
Certificate Management System	The keys, software and hardware used to verify Certificate Data, maintain a Repository, and issue and revoke Certificates.
Certificate Management Process	The policies, practices, and procedures governing the use of the Certificate Management System
Certificate Problem Report	A report of suspected Key Compromise, Certificate misuse, Certificate mis-issuance, or any other type of fraud, compromise, misuse or inappropriate conduct related to a Certificate.
ECHONET Consortium	The body that publishes the ECHONET Lite specifications, operates the associated conformance certification system, and assigns Manufacturer Codes.
ECHONET Device Certificate	A Certificate issued under this CP/CPS that identifies an individual ECHONET device and binds it to its Manufacturer and to the manufacturer-assigned product and device identifiers.

Definition	Description
ECHONET Requirements	Any requirement applicable to PKI services, certificate profiles or device identity that is published by the ECHONET Consortium and formally adopted by the PAA for the ECHONET hierarchy, as listed in Appendix D.
ECHONET Subordinate CA	A Certification Authority subordinate to the ECHONET Root CA and authorised by the PAA to issue ECHONET Device Certificates.
Hardware Crypto Module	A tamper-resistant device, with a cryptography processor, used for the specific purpose of protecting the lifecycle of cryptographic keys (generating, managing, processing, and storing).
Issuer CA	Any CA issuing Certificates under this CP/CPS
Key Compromise	A Private Key is said to be compromised if its value has been disclosed to an unauthorized person, or an unauthorized person has had access to it.
Key Pair	A Private Key and associated Public Key.
Manufacturer	The legal person that produces, or contracts the production of, an ECHONET device, and that is responsible for the accuracy of the identifiers placed in the Certificates issued to that device.
Manufacturer Code	The three-byte identifier assigned by the ECHONET Consortium to a Manufacturer, carried in ECHONET Lite property EPC 0x8A.
Private Key	The key of a Key Pair that is kept secret by the holder of the Key Pair, and that is used to create digital signatures and/or to decrypt electronic records or files that were encrypted with the corresponding Public Key.
Product Code	The identifier assigned by a Manufacturer to a product model, carried in ECHONET Lite property EPC 0x8C.
Production Number	The identifier assigned by a Manufacturer to an individual manufactured unit, carried in ECHONET Lite property EPC 0x8D.
Public Key	The key of a Key Pair that may be publicly disclosed by the holder of the corresponding Private Key and that is used by a Relying Party to verify digital signatures created with the holder's corresponding Private Key and/or to encrypt messages so that they can be decrypted only with the holder's corresponding Private Key.
Relying Party	An entity that relies upon the information contained within a Certificate.
Relying Party Agreement	An agreement which must be read and accepted by the Relying Party prior to validating, relying on or using a Certificate.
Subject Identity Information	Information that identifies the Certificate Subject, namely the Manufacturer identity and the device and product identifiers listed in section 7.1.4.
Subscriber	The entity identified as the subject in the Certificate. For ECHONET Device Certificates, the Subscriber is the Manufacturer.

Definition	Description
Subscriber Agreement	An agreement that governs the issuance and use of a Certificate that the Applicant must read and accept before receiving a Certificate.

Appendix B: CA Hierarchies

ECHONET Root

The ECHONET hierarchy is anchored in an existing IoT Root CA, which also serves other IoT device populations. It was issued on 12 August 2024 with no well-defined expiry.

Ownership and governance

The Root CA identified below is **owned by SealSQ Corp.**, which holds the certificate and the corresponding private key. This differs from the Root CAs described elsewhere in the OISTE/WISeKey Global Trust Model, which are owned by the OISTE Foundation, and the distinction is stated here so that it is not inferred from the body of this document.

Ownership of the Root does not carry authority over the policies applied beneath it. The trust model, the policies and the certification practices governing the ECHONET hierarchy are regulated by the **OISTE Foundation**, through its Policy Approval Authority, under the terms of section 1.5. In particular the PAA, and not the owner of the Root: - approves this CP/CPS and every amendment to it; - authorises each ECHONET Subordinate CA and the issuance of its certificate, under sections 3.2.2 and 3.2.6; and - conducts or commissions the assessments required by section 8, and may require a Subordinate CA to be revoked under section 4.9.1.2.

SealSQ Corp., as owner of the Root, is bound to operate it in accordance with this CP/CPS and to execute the signing operations authorised by the PAA.

Scope. This CP/CPS governs only the ECHONET Subordinate CAs listed below and the ECHONET Device Certificates issued beneath them. The Root also anchors other IoT device populations, which are outside the scope of this document and are governed by their own certificate policies and practice statements. A relying party that provisions this Root as a trust anchor therefore trusts certificate populations beyond the ECHONET one, and should constrain its validation to the Subordinate CAs and policy OIDs listed in this Appendix and in Appendix C where it intends to rely on ECHONET Device Certificates alone.

Root Information

Field	Value
Subject Name	C=CH, O=WISeKey, CN=SealSQ IOT ROOT GA
Issuer Name	C=CH, O=WISeKey, CN=SealSQ IOT ROOT GA (self-signed)
Serial Number	68:F4:11:F6:9F:A8:2A:F3:7C:7D:10:3D:28:D2:56:4C:0F:C9:D
Subject Key Identifier	C5:3B:1F:4C:A7:B9:26:D1:FD:B4:85:29:3F:74:75:C9:D3:96:C
Fingerprint (SHA-256)	4B:95:E5:F3:76:84:BE:71:45:91:F3:97:63:B7:81:56:B2:A2:C
Valid from	12 August 2024, 08:06:56 UTC
Valid until	31 December 9999, 23:59:59 UTC — no scheduled expiration
Public Key	id-ecPublicKey, 256 bit, prime256v1 (NIST P-256)
Signature Algorithm	ecdsa-with-SHA256
Key Usage	critical: digitalSignature, keyCertSign, cRLSign
Basic Constraints	critical: cA = TRUE, no pathLenConstraint
Audit scope	IoT Device Certificates, including ECHONET Device Certificates

Subordinate CA Information

The Subordinate CAs authorised to issue ECHONET Device Certificates under this Root are listed below. This table is maintained by the PAA and republished whenever a Subordinate CA is added or revoked. Subordinate CAs operated by a Manufacturer are marked as such and are technically constrained under section 7.1.5.

Subject Name	Fingerprint (SHA-256)	Operator	Allowed usage
<<<< TO BE COMPLETED AFTER THE CA CEREMONY >>>>	<<<< TO BE COMPLETED >>>>	WiSeKey	ECHONET Device Certificates

Appendix C: OID Inventory

OWGTM defines the following OID schema to identify the Certificate Policies issued under the ECHONET hierarchy.

PUBLIC-ARCH = 2.16.756.5.14

PUBLIC-ARCH.8 – Policy qualifiers for special purposes - 8.2 – Device certificates - 8.2.1 – CertifyID Device Certificate

PUBLIC-ARCH.8.3 – ECHONET Certificate Policies - 8.3.0 – OISTE/WiSeKey ECHONET CP/CPS (this document) - 8.3.1 – ECHONET Root CA CP - 8.3.2 – ECHONET Subordinate CA CP - 8.3.3 – ECHONET Device Certificate CP

The OID identifying this document is PUBLIC-ARCH.8.3.0, that is 2.16.756.5.14.8.3.0. This identifier is stable and does not carry the version number of the document: every version of this CP/CPS is identified by the same OID, and versions are distinguished by the version number stated in section 1.2 and in the revision table. The version is therefore not appended to the OID, which would otherwise collide with the policy arcs 8.3.1 to 8.3.3 defined above.

Appendix D: Adopted ECHONET Requirements

This appendix lists the documents published by the ECHONET Consortium that the PAA has formally adopted as “ECHONET Requirements” for the ECHONET hierarchy, in the sense of Appendix A and section 2.2.1. Where such a requirement conflicts with the body of this CP/CPS, the requirement prevails and the PAA shall amend this document at the next revision.

Document	Version	Date adopted by the PAA	Provisions adopted
ECHONET Lite Web API Guidelines – API specifications	1.1.4	2026-09-16	Section 5.1 (use of HTTPS for ECHONET Lite Web API endpoints)

Where this table is empty for a given matter, no ECHONET Requirement governs it and the stipulations of this CP/CPS apply in full.